WHEN MORE PREDICTIONS COVER LESS: ACCURACY AND COVERAGE UNDER RATIO RISK

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Paper under double-blind review

ABSTRACT

Selective prediction often reports how many cases it accepts, even when its risk denominator measures unequal exposure. These two choices of coverage can favor different predictions: accurate high-exposure cases supply error budget that can pay for many inaccurate low-exposure cases. We quantify this mechanism with sharp worst-case gaps, a joint bound for positive bounded exposure, and a three-context necessity result. A fixed multilabel classifier provides an exact example. Retail data supply accepted-set budget audits and a one-time external test of a previously selected policy contrast. Improving the predictor can increase both coverages without resolving their objective mismatch. In an additional prespecified urban-mobility test, exposure-based selection sacrifices 13.34 case-coverage points for 5.01 exposure points; both point risks satisfy the common cap, but their union does not. Yeast and Delicious checks do not establish a corresponding learned-policy reversal. The contribution is a quantitative account and practical audit of coverage objectives, not a new uncertainty estimator or a universal preference for exposure over cases.

1 INTRODUCTION

A selective prediction system must specify both its risk constraint and the value of acceptance. Counting accepted cases gives each case equal value. Counting their exposure, opportunity, or demand mass gives cases different weights. When risk is itself a ratio of accepted loss to accepted weight, the choice of coverage utility becomes part of the learning problem.

Case coverage alone is insufficient evidence of service to a denominator-weighted population: it can increase while covered exposure falls. This does not make equal value per case illegitimate; it makes the choice consequential. The threshold rankings follow established conditional-risk optimization. Our contribution is to quantify their disagreement: how large it can be, which settings exclude it, and how it can be measured under a common cap. A three-context family makes their case-coverage and mass-coverage gaps almost maximal, even with exact conditional information. The construction identifies the needed heterogeneity; a constant conditional denominator, for example, makes the coverage objectives identical. The acceptance unit matters: a fixed multi-label classifier admits a micro-precision reversal when the whole example is accepted or rejected. Ordinary binary precision with an unrestricted gate has a common optimum for both coverage objectives.

Retail WAPE makes the mechanism concrete. A well-predicted high-volume series creates pooled error slack that pays for many poor low-volume predictions. The added rows can have near-unit relative error yet negligible budget cost. Our experimental suite, CENSORCAST, supplies forecasts and selectors for measuring this mechanism, followed by a first item-holdout comparison and a separately frozen external comparison of two utility policies. It is an evaluation suite, not a new prediction architecture. We then ask whether improving the point predictor removes the mismatch, using newly fitted selectors for a stronger residual learner and a frozen Chronos-2 configuration. The resulting question is whether better predictions also resolve the choice of what acceptance should reward.

2 RELATED WORK AND SCOPE

Count-data evaluation. Kolassa (2016) explains why absolute-error measures, including weighted MAPE, can be unsuitable for retail count forecasts. Absolute loss elicits a median; when $\Pr(Y = 0) \geq 1/2$, zero is a median. Our new object is the acceptance budget: a poor relative-error row may still be inexpensive to accept. M5 established strong tabular forecasting baselines (Makridakis et al., 2022); we use LightGBM (Ke et al., 2017). The mechanically censored retail experiment preserves complete targets for evaluation and isolates heterogeneous forecast errors. It is one measurement setting, not a premise of the theory. Natural-stockout identification diagnostics are retained in Appendix D.4.

Ratio metrics and selective prediction. Optimization of nondecomposable and ratio-of-linear classification metrics is well established (Dinkelbach, 1967; Koyejo et al., 2014; Narasimhan et al., 2015; Bao & Sugiyama, 2020). Those works optimize prediction performance under the chosen metric. We fix the predictor and compare acceptance utilities under the same ratio constraint, asking how much case coverage can disagree with covered denominator mass.

Selection and risk control. Case coverage and its risk trade-off are foundational to selective classification (El-Yaniv & Wiener, 2010; Geifman & El-Yaniv, 2017). Conditional-loss thresholding is established in reject-option prediction (Franc et al., 2023), as are group concerns in selective regression (Shah et al., 2022). Jones et al. (2021) show that selective classification can magnify group disparities. Fisch et al. (2022) study calibration within the selected population; our distinct question is which population mass coverage rewards. Learn then Test and conformal risk control address calibration under explicit assumptions (Angelopoulos et al., 2021; 2024). Joshi et al. (2026) maximize agreement with a baseline decision policy under a chance constraint, ranking contexts by the reduction in conditional violation risk from a fallback action. Our action is forecast acceptance, the constraint is signed expected excess $\mathbb{E}[a(e - r\mu)] \leq 0$, and the utility can weight either rows or demand. Accurate high-volume cases can therefore subsidize inaccurate low-volume cases within an accepted set. Conditional thresholding remains the common mathematical tool; our analysis concerns this cross-case budget mechanism and its measurable consequences. Calibration is treated separately.

3 THE ACCEPTANCE CONTRACT

3.1 UTILITY MISMATCH UNDER RATIO RISK

Let X be the information available when acceptance is chosen. For nonnegative integrable loss L and exposure W, define

$$\ell(X) = \mathbb{E}[L \mid X], \quad w(X) = \mathbb{E}[W \mid X], \quad T = \mathbb{E}W > 0, \quad a(X) \in [0, 1].$$

Row coverage, exposure coverage, and accepted ratio risk are

$$c(a) = \mathbb{E}a, \qquad d_W(a) = \frac{\mathbb{E}[aW]}{T}, \qquad R(a) = \frac{\mathbb{E}[aL]}{\mathbb{E}[aW]}. \tag{1}$$

The accepted denominator must be positive. The contract is $R(a) \leq r$ and $c(a) \geq c_0$. Its signed excess is

$$g_r = L - rW, \qquad m_r = \ell - rw, \qquad R(a) \leq r \iff \mathbb{E}[am_r] \leq 0. \tag{2}$$

WAPE is the specialization $L = |Y - f(X)|$, $W = Y$, with $Y, f \geq 0$; then $\ell = e$, $w = \mu$, and $d_W = d$ is demand coverage.

Fixed-row-coverage excess is minimized by accepting the smallest m_r values, with randomization on ties (Proposition 1 in Appendix A). To maximize exposure instead, change measure to $d\nu = w\, dP_X/T$ after rejecting $w = 0, \ell > 0$ contexts: then $d_W(a) = \mathbb{E}_\nu a$ and $R(a) = \mathbb{E}_\nu[a\ell/w]/\mathbb{E}_\nu a$.

Proposition 2 (Exposure-weighted ranking). *If a positive-exposure feasible rule exists and the row floor is nonbinding, exposure coverage is optimized by thresholding ℓ/w, with boundary randomization. Reject $w = 0, \ell > 0$ contexts; $w = \ell = 0$ contexts are neutral for this objective. The ranking is independent of r; its threshold is not.*

Thus the population rankings are $\ell - rw$ for rows and ℓ/w for exposure. A fixed ranking's largest feasible threshold maximizes either coverage because its acceptance sets are nested; objective comparisons must therefore also compare ranking families (Appendix C).

For disjoint context sets K, U, write

$$M_A = \mathbb{E}[\mathbf{1}_A w], \qquad B = -\mathbb{E}[\mathbf{1}_K m_r], \qquad M_K > 0, \quad B \geq 0.$$

Here B is the excess slack supplied by a common accepted set.

Proposition 3 (Sharp objective reversal for ratio risk)**.** *Fix $r > 0$, $\rho > r$, and $\varepsilon > 0$. If $\ell = \rho w$ on U and $0 < M_U \leq \varepsilon \Pr(U)$, then $R(\mathbf{1}_U) = \rho$ and $K \cup U$ satisfies the cap exactly when $(\rho - r) M_U \leq B$. Accepting U adds $\Pr(U)$ row coverage but at most $\varepsilon \Pr(U)/T$ exposure coverage.*

For every $0 \leq c_0 < 1$ and $\eta > 0$, there exists a three-context law with $w > 0$, $0 \leq \ell \leq \rho w$, and exact conditional moments whose unique row- and exposure-optimal policies a_R, a_W satisfy the cap and the nonbinding row floor, and obey

$$c(a_R) - c(a_W) > 1 - c_0 - \eta, \qquad d_W(a_W) - d_W(a_R) > 1 - \eta.$$

The universal bounds $1 - c_0$ and 1 are jointly sharp. For $0 < r < 1$ and $\rho = 1$, this sharpness also holds with binary $0 \leq L \leq W \leq 1$.

The sharpness statement ranges over joint laws that can realize the conditional moments in Appendix A; it is not a claim about every fixed metric class. If W is constant, for example, $d_W = c$ and reversal is impossible. For a fixed multi-label classifier, whole-example abstention yields an exact micro-precision reversal (Appendix A.2). Ordinary binary precision with an unrestricted gate instead has a common optimum for both coverage objectives. For whole-example micro-precision with $1 \leq w \leq m$, each coverage gap is at most $(m-1)/(m+1)$; Appendix A.2 sharpens this to a joint bound.

When reversal is limited or absent. For row- and exposure-optimal policies on a common feasible set, let $x = c(a_R) - c(a_W) \geq 0$ and $y = d_W(a_W) - d_W(a_R) \geq 0$. If $0 < u \leq w \leq v$, then, with $D = (v-u)/(v+u)$,

$$y \leq \frac{D - x}{1 - Dx}, \qquad 0 \leq x, y \leq D. \tag{3}$$

Thus constant positive conditional exposure rules out reversal. Moreover, with unrestricted randomized acceptance under just the pooled risk cap and row floor, at most two positive-probability, positive-exposure contexts have a common optimum whenever feasible. Strict reversal requires at least three such contexts. Appendix A.2 proves both claims; variation alone is not sufficient, as $\ell \equiv 0$ makes accept-all optimal for any exposure.

WAPE corollary and estimation. For $0 < r < 1$, setting $\rho = 1$, $W = Y$, and $f = 0$ on U gives $e = \mu$: near-zero-demand rows have budget cost $(1-r)\mu$ yet unit row reward. The three-context law is realizable by deterministic nonnegative forecasts (Appendix A), so both sharp bounds hold for WAPE. A low-exposure share alone cannot bound the forgone exposure without a positive excess-rate margin; the appendix gives that bound and the exact budget identity. More generally $|e - \mu| \leq f$. Exact-head optimality does not imply estimated-ratio robustness. Section 5.3 and Appendix A distinguish objective alignment from small-denominator error amplification.

3.2 A FIXED CLASSIFIER: EXAMPLES VERSUS POSITIVE PREDICTIONS

Consider a fixed two-label classifier with whole-example abstention. Exposure W counts its positive predictions, and loss L counts false positives, so $1 - R$ is accepted micro-precision. Three contexts have the following known conditional moments:

	K	U	V
$\Pr(X)$	$2/5$	$2/5$	$1/5$
$h(X)$	$(1,0)$	$(1,0)$	$(1,1)$
$w(X)$	1	1	2
$\ell(X)$	0	$3/10$	$1/2$

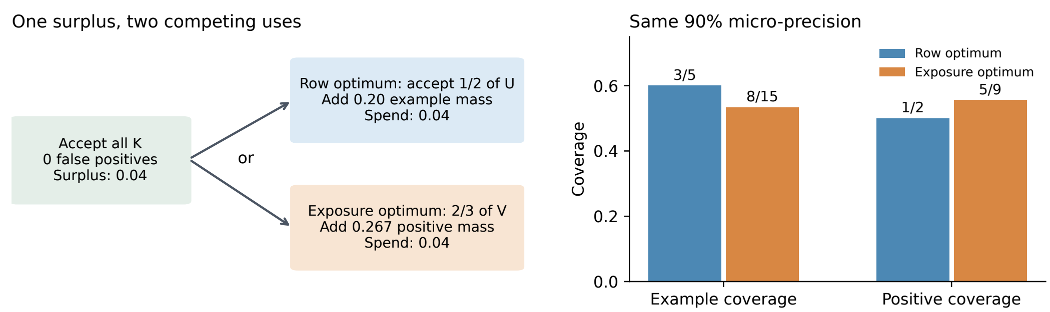

Figure 1: Exact fixed-classifier example, not an empirical benchmark. Left: the accurate context K supplies 0.04 expected excess units; either competing allocation uses all of them. Right: the same risk budget buys more examples or more positive predictions, but not both. Fractional acceptance is independent randomization within a context.

At risk cap $r = 1/10$ and case floor $c_0 = 7/20$, every optimum accepts K. Its risk surplus funds the remaining constraint $4a_U + 3a_V \leq 2$. Maximizing accepted examples chooses $(a_K, a_U, a_V) = (1, 1/2, 0)$, giving $(c, d_W) = (3/5, 1/2)$. Maximizing accepted positive predictions instead chooses $(1, 0, 2/3)$, giving $(8/15, 5/9)$. Both attain exactly 90% micro-precision. The classifier is unchanged; only the value assigned to acceptance differs (Figure 1). Appendix A.2 specifies the label distribution and verifies both optima.

For ordinary binary precision, accepting all predicted-negative cases costs no risk budget, and completed gates satisfy $c = 1 - T + T d_W$. Both objectives then have a common optimum. This contrast makes the acceptance unit consequential: gating each label separately would change the multi-label problem above.

Scope under censoring. The mismatch theorem does not require censored observations. Its inputs are the conditional loss and exposure of a fixed predictor. Mechanical censoring lets us retain the complete target to estimate and audit these inputs; natural stockouts generally do not. Observed sales alone cannot certify a cap below one when compatible demand tails are unbounded (Appendix A.3). This is a limit on identifying the theorem's inputs, not an additional explanation of objective reversal.

3.3 Calibration is a separate claim

Independent calibration clusters and known bounded cluster losses permit simultaneous finite-family concentration checks; the supplement states a standard Hoeffding construction (Appendix A.3). Our retail counts lack a justified population bound and share store/calendar shocks. Accordingly, development intervals are descriptive; the held-out comparisons use prespecified item-bootstrap approximations, not that theorem.

4 Method and experimental design

4.1 Data, chronology, and the audit boundary

The M5 split has three disjoint item partitions, with all ten stores of an item kept together. *Development* has 1,829 items (18,290 series) and supplies fitting and threshold selection. The first *item holdout*, called *guardian* in the result files, has 610 items: 42 policies were frozen before its one-time comparison. Its saved predictions subsequently supported exploratory diagnostics. The final *external* partition has 610 other items and is used once for two policies fixed after that exploration (Section 5.5). The first holdout shares development dates and stores; the external primary test uses later forecast origins, still in the same stores. Recorded M5 unit sales are the target Y.

After 56 uncensored warmup days, mechanical censoring depends on previous observed sales:

$$L_t = 0.90L_{t-1} + 0.10S_{t-1}, \quad C_t = \max\{1, \lceil 1.15L_t + 0.25\sqrt{L_t + 1}\rceil\}, \quad S_t = \min(Y_t, C_t). \tag{4}$$

The previously fixed process hides 30.42% of recorded demand mass and censors 12.67% of development rows on days 1314–1913. Initialization is specified in Appendix B.

Forecast horizons are one through seven days from weekly origins. Chronological blocks separately fit the predictor, select its configuration, train the risk heads, and choose thresholds using two calibration windows, A and B. The subsequent development evaluation uses days 1794–1913; origin-availability filtering leaves 2,085,060 rows. Appendix Table 3 gives every date boundary and the corresponding first-holdout windows. All features use information available at the forecast origin.

4.2 Matched point forecasts and censor-adapter controls

The original experiment compares observed-sales L1 and Poisson forecasts, rescaling, and a gated censoring adapter. Its point predictor is fixed before selector comparison. Appendix G.5 preserves the model specifications and adapter equation; the accuracy intervention uses separate predictors and newly fitted selectors.

4.3 Matched selectors and operational checks

Risk learners share 600,000 sampled risk-training rows, 21 origin-valid features, 220 trees, 31 leaves, and three seeds. Squared-loss heads estimate conditional error $\hat{e}$, demand $\hat{\mu}$, or excess. The two-head score is $\hat{e} - r\hat{\mu}$ (440 total trees); the matched-budget control uses the first 110 trees of each head. Nonnegative heads are clipped at zero. Forecast-normalized error is $\hat{e}/\max(f, 0.25)$; demand-normalized error is $\hat{e}/\max(\hat{\mu}, 0.25)$. These names are used throughout.

We require WAPE at most $r = 0.6401298$ and row coverage at least 0.35 in both calibration blocks. Each score accepts a ranked prefix. We search all distinct thresholds and choose the feasible threshold with the largest minimum A/B row coverage. Category-specific thresholds follow the same rule; an infeasible category is fully rejected.

The cap is a 15% reduction from the reference Poisson model's WAPE, 0.7530939208, on all 2,194,800 development point-selection rows (days 1314–1433, before origin purging). The 0.35 row floor retains a five-point buffer above the original 0.30 minimum; it is a study setting, not a retail standard.

The original holdout comparison uses frozen policies; subsequent analyses reuse saved predictions and are post-hoc. Thresholds and families are selected on development calibration. Item bootstrap keeps each item's stores and days together and conditions on fitted models. Complete grids and access records remain in the appendix.

5 Results

5.1 More accepted rows can mean less covered demand

For a common accepted set K and disjoint added set U, let M_B and E_B be the demand and absolute-error mass of set B. Excess is additive: $(E_{K\cup U} - rM_{K\cup U}) = (E_K - rM_K) + (E_U - rM_U)$. When both masses are positive, the combined set meets the cap exactly when

$$M_U(R_U - r) \le M_K(r - R_K). \tag{5}$$

Thus many inaccurate low-volume rows can consume little of the common set's available error budget. We measure each term through an accepted-set decomposition.

On the previously evaluated development window, direct excess gains 6.08 case-coverage points over forecast-normalized error but loses 5.71 demand-coverage points. Their common accepted set has WAPE 0.5958. The additional direct-only rows have WAPE 1.0026 and mean forecast 0.039 units, yet consume just 61.88% of that common set's surplus. The competing exclusive set carries

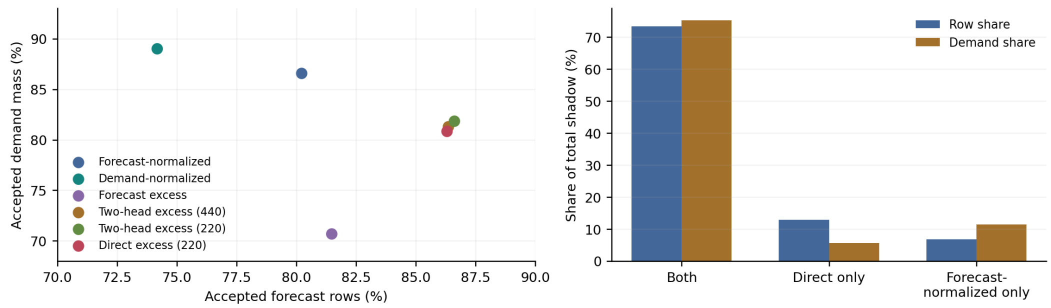

Figure 2: Strong forecaster. Left: three-seed means under the same row-coverage threshold selection; row and demand coverage disagree. Right: first-seed direct-excess versus forecast-normalized-error acceptance sets. The extra direct-only set contains more rows but less demand than the relative-only set. Both panels describe the previously evaluated development window.

roughly twice as much demand while spending 52.01% of the surplus. Thus either policy can satisfy the pooled cap while making a different choice of what to cover. Appendix C.1 gives the complete masses and row counts.

5.2 Transfer to previously held-out items

On the 610-item first holdout, three frozen fits reproduce the same direction: direct excess gains 6.49 case-coverage points over forecast-normalized error while losing 4.36 demand-coverage points. This first comparison also includes a direct-versus-two-head architecture control; neither architecture establishes a coverage advantage independent of risk (Appendix C.1).

The joint operating check fails: all 24 adjusted risk upper bounds exceed the cap, although the 24 coverage lower bounds pass. Hobbies and Household fail even at the later-window point estimates; pooled risk also fails in calibration-period B. These failures distinguish transfer of a utility contrast from transfer of a risk guarantee. Appendix B.2 preserves the complete endpoint family and access record.

Group budgets and certification. Pooled point-risk feasibility is separate from simultaneous or groupwise certification. Hobbies and Household exhibit point-risk violations, while failure of an adjusted pooled upper bound means a guarantee was not established. Error-information substitutions, shared-head feature interventions, and a group-budget LP delimit the tested policy family's shortcomings (Appendix C). These checks support reporting group budgets, not claiming that all selective policies must fail.

5.3 Estimated ratios require regularization disclosure

Changing utility on a fixed nested ranking leaves its largest feasible threshold unchanged; different utilities require comparing rankings. The demand-selected family uses $\hat{e}/\max(\hat{\mu}, 0.25)$. Removing that floor reverses its first-holdout comparison with forecast normalization (Appendix C). Exact-ratio optimality does not order fitted scores: $\partial(e/\mu)/\partial\mu = -e/\mu^2$, whereas $\partial(e - r\mu)/\partial\mu = -r$. A floor or positive row weight limits small-denominator amplification, a possible source of instability rather than an identified explanation of any particular empirical reversal.

5.4 A constructive path between coverage objectives

Let $U_\lambda = \lambda c + (1-\lambda)d$, $T = \mathbb{E}Y$, and $w_\lambda = \lambda + (1-\lambda)\mu/T$. For a nonbinding row floor, the same weighted-measure argument orders

$$s_\lambda = \frac{e - r\mu}{\lambda + (1-\lambda)\mu/T}, \qquad 0 \le \lambda \le 1. \tag{6}$$

It recovers excess at one endpoint and demand-ratio ordering at the other. We evaluate the fixed grid $\{0, 0.25, 0.5, 0.75, 1\}$ with the same first-seed heads, no additional max-denominator regularization, and a common development-only estimate of T. All thresholds are chosen on development before application to previously evaluated first-holdout predictions (Appendix C.4).

Appendix Table 12 reports the full development A/B path. On the item holdout, at $\lambda = 0.25$, later-window row/demand coverage is 84.31%/88.67% with WAPE 0.6272; at the composed-excess endpoint $\lambda = 1$, it is 87.46%/83.78% with WAPE 0.6296. This offers an explicit utility choice. The transferred points need not form an empirical Pareto frontier, and all five still fail period-B pooled WAPE. No mixture is promoted to a confirmed operating policy.

Table 1: One-time external comparison: 610 items, days 1919–1941. Two policies were fixed before opening; thresholds and heads are unchanged. Point WAPE passes the pooled cap, but the 48-endpoint operating check fails.

Fixed policy	Rows (%)	Demand (%)	WAPE
Row utility, $\lambda = 1$	85.92	83.19	0.6065
Mixed utility, $\lambda = 0.25$	82.57	88.35	0.6081

Only the first seed and $\lambda \in \{1, 0.25\}$ were frozen for this test. The pair was selected after the first holdout; other external seeds and the full λ grid were not evaluated. The test verifies transfer of this selected contrast.

5.5 A ONE-TIME EXTERNAL CONFIRMATION OF THE FIXED CONTRAST

After inspecting the first-holdout path, we replaced an earlier pending policy pair with $\lambda = 1$ and $\lambda = 0.25$, then froze their thresholds and first-seed heads before external access (Appendix B.3). The 610 unseen items contribute 140,300 rows on days 1919–1941; all forecast origins follow the last previously evaluated development day. Mixed utility gains 5.16 demand-coverage points and loses 3.35 row points (Table 1). Paired one-sided 97.5% item-bootstrap bounds are $+3.17$ points for the demand difference's lower bound and -2.48 for the row difference's upper bound. Both prespecified directions pass. The test concerns this selected learned-policy contrast; it does not test the population theorem, pure-demand optimality, or the full utility path. Pooled point risks pass the cap, but all 24 multiplicity-adjusted WAPE upper bounds exceed it; all 24 row lower bounds pass. Failure to certify pooled risk is distinct from the Hobbies/Household point-risk violations. The adjusted bounds do not establish that every underlying risk exceeds the cap. A descriptive decomposition of the same saved masks links the contrast to equation (5): each exclusive set spends about 30% of the common slack, while the row-only set has WAPE 1.0046 and carries much less demand (Appendix B.3). The two exclusive excesses sum to less than the common slack, so their union is also point-feasible on this external sample. The external test therefore confirms transfer of a selected contrast, not transfer of binding budget competition or Pareto optimality.

5.6 DOES BETTER FULL-COVERAGE ACCURACY REMOVE THE MISMATCH?

Better forecasts can expand useful acceptance, but this alone does not specify how to spend the error budget. If a predictor's full-coverage risk is at most r, accepting all cases is a common optimum and mismatch disappears. Full-coverage risk above the cap is necessary for a strict objective reversal, but it is not sufficient: the same selective rule can still optimize both utilities (Appendix F.1). We test empirically whether reversal persists for the improved predictors here.

We compare a hierarchy-feature L1 forecaster with category scaling, its forward-time learned residual correction, and one fixed univariate Chronos-2 configuration (Ansari et al., 2025). All are evaluated on the same previously evaluated development rows. We fit a separate error head for each predictor and a common demand head, then calibrate and evaluate on shared dates (Appendix E). Chronos weights remain frozen; its selector is task-trained. This tests acceptance within each pipeline, not training-matched backbone superiority.

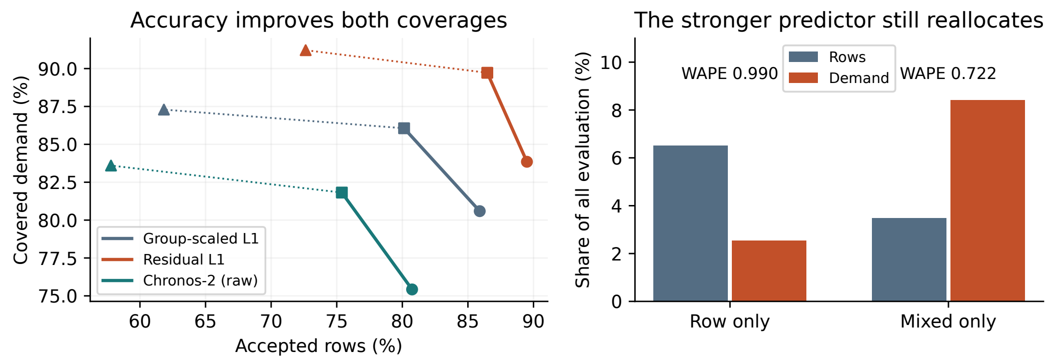

Figure 3: New matched-date comparison. Left: moving to a stronger predictor raises both coverages, while changing utility within a predictor trades them off. Circles, squares, and triangles denote $\lambda = 1, 0.25, 0$. Lines connect evaluated settings, not Pareto frontiers. Right: the improved predictor's exclusive acceptance sets retain the budget mechanism.

Table 2: New retrospective comparison on the same 2,085,060 M5 development evaluation rows. Each predictor has a newly fitted error head; all share a newly fitted demand head, head-training rows, cap, and calibration dates. Higher coverage is better; lower WAPE is better. Full WAPE uses every row. These are fitted-policy outcomes, not population optima or a new external test.

Predictor	Full WAPE	λ	Rows (%)	Demand (%)	Accepted WAPE
Group-scaled L1	0.67139	1	85.90	80.59	0.63272
Group-scaled L1	0.67139	0.25	80.13	86.06	0.63039
Residual L1	0.65956	1	89.50	83.85	0.63298
Residual L1	0.65956	0.25	86.48	89.72	0.63116
Chronos-2 (raw)	0.68310	1	80.74	75.42	0.63568
Chronos-2 (raw)	0.68310	0.25	75.39	81.82	0.63007

The residual learner lowers full-coverage WAPE by 1.76% relative to category-scaled L1. Both its row-weighted and mixed policies increase *both* coverages over the corresponding base policies. Yet switching utility within that improved predictor still gains 5.87 demand-coverage points while losing 3.02 row points. The paired item-bootstrap intervals are $[4.84, 6.88]$ and $[-3.57, -2.43]$ points, respectively. The same signs occur for the base and Chronos predictors, and for all three pure-demand endpoints (Appendix Table 17). These are retrospective, fit-conditional comparisons.

A constructive improvement is already visible: the improved predictor at $\lambda = .25$ exceeds the base at $\lambda = 1$ in both coverages (86.48/89.72 versus 85.90/80.59 percent). Better prediction and a changed utility can therefore improve both reported quantities. This cross-predictor comparison is retrospective, not a transferred guarantee. Budget accounting connects the result to the theorem. For the improved predictor, the common set supplies 43,525.76 units of slack. The row-only set spends 26,046.82 (59.84% of that slack), accepting 135,756 rows with mean forecast 0.107, WAPE 0.9904, and only 2.55% of total demand. The mixed-only set instead adds 72,762 rows carrying 8.42% of demand. With the improved predictor, the row-coverage sacrifice is smaller than for the base (3.02 points, compared with 5.77 for the base), while the different allocations remain. Hobbies and Household still miss the cap under every pooled policy; this experiment establishes no groupwise deployment guarantee.

Sensitivity and simple rankings. The 288-condition development sweep preserves the mixed-versus-row contrast at strict caps. The error-only regression ranking is infeasible at the original cap; increasing the row floor either preserves a threshold or makes its family infeasible (Appendix F). Descending predicted demand covers more demand than the fitted pure-ratio score for all three predictors, while accepting fewer rows. Thus exact-information optimality is not an empirical superiority claim for the fitted ratio score (Appendix G).

5.7 A NON-RETAIL TEST WITH COMPETING BUDGET ALLOCATIONS

On the public UCI Bike Sharing data, we fix the predictors, disjoint day-group splits, primary cap rule, and selectors before evaluating held-out outcomes. This is conditional count prediction using recorded weather, not a future-weather forecast. At the primary cap $r = .13281$, row/exposure selection accepts $87.89\%/74.55\%$ of hours and covers $85.54\%/90.56\%$ of recorded rentals. Both point risks pass. Day-bootstrap intervals adjusted across the two primary dataset contrasts exclude zero in both directions (Appendix G). The common set supplies 5,219 loss units of slack; the two exclusive sets require 3,569 and 2,399. Each fits separately but their union exceeds the cap, exhibiting budget competition beyond retail.

The prespecified Delicious primary check is infeasible; Yeast's earlier check has no clear reversal. Kendall rank agreement is high on Yeast and lower on M5, a useful descriptive diagnostic rather than a sufficient condition for reversal. All declared conditions, including these negative results, remain in the supplement.

6 LIMITATIONS

The new predictor comparison is retrospective on previously evaluated M5 development items; it does not extend the independent external confirmation to new predictors. Its selectors use retained pre-censoring targets, a single fit, and chronological point constraints. Approximate item-bootstrap intervals condition on fits and do not remove shared store/calendar shocks. Exact rankings can differ from fitted rankings, and arbitrary fitted policies need not lie on a population Pareto frontier. The original joint operating checks fail. Natural-stockout diagnostics concern identification rather than coverage geometry (Appendix D.4). The bounded-exposure envelope can be vacuous when the exposure lower bound approaches zero, as in retail; we do not treat it as an informative retail error bar. The empirical coverage gaps are not estimates of the worst-case bound: its sharp construction allows conditional exposure to approach zero. The cap/floor sweep is development-only; it establishes neither seed robustness nor the full utility path on the external partition. The new mobility split is randomized by day, and its bootstrap conditions on fitted models. Forecast and covered-demand improvements are not measured profit, service-level, or inventory-cost improvements.

7 CONCLUSION

Under ratio-risk constraints, familiar threshold rules can optimize substantially different notions of coverage. We identify when case and exposure objectives can disagree, bound their disagreement, and measure the underlying budget allocation. The original external comparison confirms a selected policy contrast; stronger-forecast experiments separately examine its sensitivity to prediction quality. Forecasting systems should improve full-coverage accuracy, specify acceptance utility, and report both coverages with group-level budget audits.

AI USE STATEMENT

AI tools assisted with proofs, experimental design and implementation, analysis, and English writing and translation. The authors take full responsibility for all content and claims.

ETHICS STATEMENT

Mechanically hidden recorded sales must not be represented as identified customer demand. Pooled acceptance can concentrate errors in particular product groups; we report those failures and do not recommend deployment from these results. No human-subject intervention, customer profiling, or live replenishment action was performed.

REPRODUCIBILITY STATEMENT

The supplement supplies experiment scripts, fixed configurations, models, hashes, result records, and origin/mask tests. Appendix B gives preprocessing, dates, and parameters; Appendix A gives proofs. The source package provides separate commands for theory and frozen-object verification, statistical replay, and rebuilding the manuscript. External item sufficient statistics reproduce both primary bounds and all 48 secondary endpoints. Preparation/design source regenerates data from the cited M5 release; previously evaluated prediction caches support the exploratory audits.

The anonymous repository at `https://github.com/censorcast-review/censorcast` provides the manuscript source, experiment code, saved prediction arrays, and verification commands for the complete numerical replay.

REFERENCES

Anastasios N. Angelopoulos, Stephen Bates, Emmanuel J. Candès, Michael I. Jordan, and Lihua Lei. Learn then test: Calibrating predictive algorithms to achieve risk control, 2021. URL https://arxiv.org/abs/2110.01052.

Anastasios N. Angelopoulos, Stephen Bates, Adam Fisch, Lihua Lei, and Tal Schuster. Conformal risk control. In *International Conference on Learning Representations*, 2024. URL https://arxiv.org/abs/2208.02814.

Abdul Fatir Ansari, Oleksandr Shchur, Jaris Küken, Andreas Auer, Boran Han, Pedro Mercado, Syama Sundar Rangapuram, Huibin Shen, Lorenzo Stella, Xiyuan Zhang, Mononito Goswami, Shubham Kapoor, Danielle C. Maddix, Pablo Guerron, Tony Hu, Junming Yin, Nick Erickson, Prateek Mutalik Desai, Hao Wang, Huzefa Rangwala, George Karypis, Yuyang Wang, and Michael Bohlke-Schneider. Chronos-2: From univariate to universal forecasting. *arXiv preprint arXiv:2510.15821*, 2025. URL https://arxiv.org/abs/2510.15821.

Han Bao and Masashi Sugiyama. Calibrated surrogate maximization of linear-fractional utility in binary classification. In *Proceedings of the Twenty Third International Conference on Artificial Intelligence and Statistics*, volume 108 of *Proceedings of Machine Learning Research*, pp. 2337–2347, 2020. URL https://proceedings.mlr.press/v108/bao20a.html.

Werner Dinkelbach. On nonlinear fractional programming. *Management Science*, 13(7):492–498, 1967. doi: 10.1287/mnsc.13.7.492.

Ran El-Yaniv and Yair Wiener. On the foundations of noise-free selective classification. *Journal of Machine Learning Research*, 11(53):1605–1641, 2010. URL https://jmlr.org/papers/v11/el-yaniv10a.html.

Adam Fisch, Tommi Jaakkola, and Regina Barzilay. Calibrated selective classification. *Transactions on Machine Learning Research*, 2022. URL https://arxiv.org/abs/2208.12084.

Vojtech Franc, Daniel Prusa, and Vaclav Voracek. Optimal strategies for reject option classifiers. *Journal of Machine Learning Research*, 24(11):1–49, 2023. URL https://jmlr.org/papers/v24/21-0048.html.

Yonatan Geifman and Ran El-Yaniv. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems*, volume 30, 2017. URL https://papers.nips.cc/paper_files/paper/2017/hash/4a8423d5e91fda00bb7e46540e2b0cf1-Abstract.html.

Wassily Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963. doi: 10.1080/01621459.1963.10500830.

Erik Jones, Shiori Sagawa, Pang Wei Koh, Ananya Kumar, and Percy Liang. Selective classification can magnify disparities across groups. In *International Conference on Learning Representations*, 2021. URL https://arxiv.org/abs/2010.14134.

Sunay Joshi, Tao Wang, Hamed Hassani, and Edgar Dobriban. Risk-controlled post-processing of decision policies, 2026. URL https://arxiv.org/abs/2605.06479.

Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

Stephan Kolassa. Evaluating predictive count data distributions in retail sales forecasting. *International Journal of Forecasting*, 32(3):788–803, 2016. doi: 10.1016/j.ijforecast.2015.12.004.

Oluwasanmi Koyejo, Nagarajan Natarajan, Pradeep Ravikumar, and Inderjit S. Dhillon. Consistent binary classification with generalized performance metrics. In *Advances in Neural Information Processing Systems*, volume 27, 2014.

Spyros Makridakis, Evangelos Spiliotis, and Vassilios Assimakopoulos. M5 accuracy competition: Results, findings, and conclusions. *International Journal of Forecasting*, 38(4):1346–1364, 2022. doi: 10.1016/j.ijforecast.2021.11.013.

Harikrishna Narasimhan, Harish Ramaswamy, Aadirupa Saha, and Shivani Agarwal. Consistent multiclass algorithms for complex performance measures. In *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *Proceedings of Machine Learning Research*, pp. 2398–2407, 2015. URL https://proceedings.mlr.press/v37/narasimhanb15.html.

Abhin Shah, Yuheng Bu, Joshua K. Lee, Subhro Das, Rameswar Panda, Prasanna Sattigeri, and Gregory W. Wornell. Selective regression under fairness criteria. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pp. 19598–19615, 2022. URL https://proceedings.mlr.press/v162/shah22a.html.

University of Nicosia. M5 forecasting accuracy. Zenodo dataset record, 2020. URL https://zenodo.org/records/10203108.

A Proofs, identification, and calibration

A.1 Fixed row coverage and exposure-weighted coverage

Proposition 1 (Fixed-coverage threshold principle)**.** *Assume $\mathbb{E}|L - rW| < \infty$. Among rules with $\mathbb{E}a = c$, accepting the smallest $m_r = \mathbb{E}[L - rW \mid X]$ values minimizes expected excess. Boundary randomization resolves atoms. The minimum excess is $\int_0^c Q_m(u)\,du$, where Q_m is the quantile function of m_r. A ratio-risk assertion additionally requires positive accepted exposure.*

Choose t with $\Pr(m_r < t) \le c \le \Pr(m_r \le t)$, and let a^* accept below t, reject above it, and randomize on the tie to attain c. For any other rule of the same coverage,

$$(a - a^*)(m_r - t) \ge 0, \qquad \mathbb{E}[a(L - rW)] - \mathbb{E}[a^*(L - rW)] = \mathbb{E}[(a - a^*)(m_r - t)] \ge 0.$$

This proves the threshold rule and the lower-quantile expression. The limiting rules cover $c = 0, 1$. This minimizes excess, not the ratio itself at fixed row coverage, because accepted exposure can vary.

For Proposition 2, $w = 0$ implies $W = 0$ almost surely conditionally. These contexts add no exposure: reject them if $\ell > 0$; otherwise they are neutral. On $\{w > 0\}$, set $d\nu = w\, dP_X/T$ and $\xi = \ell/w$. Then

$$\mathbb{E}[aW] = T\mathbb{E}_\nu a, \qquad \mathbb{E}[aL] = T\mathbb{E}_\nu[a\xi].$$

The exchange argument under ν minimizes the numerator at fixed exposure coverage by ordering ξ. Maximizing feasible exposure coverage thus admits an ℓ/w threshold when a positive-exposure feasible rule exists. An additional binding row floor can change that optimizer. Randomization handles atoms; the ranking is independent of r, while the selected threshold depends on r.

A.2 Sharp reversal for nonnegative ratio risks

Proof of Proposition 3. Since $\ell = \rho w$ on U, its error-to-exposure ratio is ρ and its excess mass is $(\rho - r)M_U$. The combined excess is $-B + (\rho - r)M_U$, proving the first assertion and the coverage increments.

For sharpness, take $k, p, q > 0$ with $k + p + q = 1$, $k > c_0$, $q > p$, and $0 < \varepsilon < p/q$. Set the conditional moments to

	K	U	V
$\Pr(X)$	k	q	p
w	$\dfrac{q(\rho-r)\varepsilon}{rk}$	ε	1
ℓ	0	$\rho\varepsilon$	$r + \dfrac{q}{p}(\rho - r)\varepsilon$

They can be realized simply by deterministic $(L, W) = (\ell, w)$. Their excesses are

$$m_K = -\frac{q(\rho - r)\varepsilon}{k}, \quad m_U = (\rho - r)\varepsilon, \quad m_V = \frac{q}{p}(\rho - r)\varepsilon.$$

Every optimum accepts K, since doing so increases either utility and relaxes the constraint. Its budget is $B = q(\rho - r)\varepsilon$. Once K is accepted, feasibility reduces to $a_U + a_V \le 1$. Row utility is maximized uniquely at $(a_U, a_V) = (1, 0)$ since $q > p$; exposure utility is maximized uniquely at $(0, 1)$ since $q\varepsilon < p$. These arguments optimize over all fractional acceptance rules. Both policies have ratio risk r and row coverage above c_0. With $T = p + q\rho\varepsilon/r$,

$$\Delta c = q - p, \qquad d_W(a_R) = \frac{q\rho\varepsilon}{rp + q\rho\varepsilon}, \qquad \Delta d_W = \frac{r(p - q\varepsilon)}{rp + q\rho\varepsilon}.$$

Take $k > c_0$ sufficiently near c_0 and then $p > 0$ sufficiently small that $q - p = 1 - k - 2p > 1 - c_0 - \eta$; for these masses, $\varepsilon \downarrow 0$ gives $\Delta d_W \to 1$. The opposite inequalities follow from $c(a_R) \le 1$, $c(a_W) \ge c_0$, and $0 \le d_W \le 1$.

For $0 < r < 1$, choose $\rho = 1$ and scale all moments by $h = 1/\max(1, w_K)$, which leaves risks, both coverages, and both optimizers unchanged. Then $0 \le \ell \le w \le 1$. At each context, put probabilities $\ell, w - \ell, 1 - w$ on $(L, W) = (1, 1), (0, 1), (0, 0)$. This realizes the same conditional moments with nested binary events, proving bounded-loss sharpness. For the deterministic WAPE specialization before scaling, take $Y = (w_K, \varepsilon, 1)$ and $f = (w_K, 0, (1 - r)(1 - q\varepsilon/p))$. These nonnegative forecasts realize precisely the $\rho = 1$ moments. □

Scope of the realizability condition. The displayed moment family is a sufficient condition for a fixed loss/exposure model class to inherit the sharp bounds. In this three-context family the strict inequalities $q > p$ and $q\varepsilon < p$ are also necessary for these two distinct, unique optima. They are not necessary conditions for reversal in arbitrary larger classes. Ratio form, nonnegativity, and integrability alone do not make every specified metric class rich enough: constant w makes exposure and row utility proportional. For a precision-like error ratio, $L = \mathbf{1}\{\widehat{Y} = 1, Y = 0\}$ and $W = \mathbf{1}\{\widehat{Y} = 1\}$; for a recall-like miss ratio, $L = \mathbf{1}\{Y = 1, \widehat{Y} = 0\}$ and $W = \mathbf{1}\{Y = 1\}$. Both have $L \le W$, but realization requires acceptance to be chosen using information for which the joint event probabilities can take the three specified values. A deterministic prediction included in X restricts these moments. Recall here is conditional on accepted cases; a metric counting rejected positives in its fixed denominator is a different constraint.

A margin-dependent bound. For alternatives $K \cup U$ and $K \cup V$, suppose $\ell = \rho w$ on U and U exhausts the common slack: $B = (\rho - r)M_U > 0$. If $K \cup V$ satisfies the cap, $M_V > 0$, and $R(\mathbf{1}_V) - r \geq \delta > 0$, then

$$d_W(\mathbf{1}_{K\cup V}) - d_W(\mathbf{1}_{K\cup U}) \leq \frac{M_U}{T}\left(\frac{\rho - r}{\delta} - 1\right).$$

Indeed, $\delta M_V \leq \mathbb{E}[\mathbf{1}_V m_r] \leq B$. If both alternatives exhaust the same budget, the exact identity is

$$\Delta d_W = \frac{M_U}{T}\frac{\rho - R(\mathbf{1}_V)}{R(\mathbf{1}_V) - r}.$$

The construction has $M_U/T \to 0$ but $R(\mathbf{1}_V) - r \to 0$ as well; this is why exposure share alone gives no vanishing bound. For WAPE near-zero forecasts satisfy $|\mathbb{E}[\mathbf{1}_U e] - M_U| \leq \mathbb{E}[\mathbf{1}_U f]$, and hence $|R(\mathbf{1}_U) - 1| \leq \mathbb{E}[\mathbf{1}_U f]/M_U$.

Finite perturbations of estimated rankings. Let $\widehat{\ell} = \ell + \delta_\ell$, $\widehat{w} = w + \delta_w$, where $w > 0$ and $|\delta_w| \leq \beta w$ for some $0 \leq \beta < 1$. Direct subtraction gives

$$\left|\frac{\widehat{\ell}}{\widehat{w}} - \frac{\ell}{w}\right| = \frac{|w\delta_\ell - \ell\delta_w|}{w|w + \delta_w|} \leq \frac{|\delta_\ell| + (\ell/w)|\delta_w|}{(1-\beta)w}.$$

In contrast, $|\widehat{m}_r - m_r| \leq |\delta_\ell| + r|\delta_w|$. The local derivatives with respect to w are $-\ell/w^2$ and $-r$; the ratio's sensitivity with respect to ℓ is $1/w$. These are sensitivity statements, not a proof that denominator noise alone causes the observed empirical ordering. For fixed $T > 0$, write $b = (1-\lambda)/T$ and $s_\lambda = (\ell - rw)/(\lambda + bw)$. When $0 < \lambda \leq 1$ and $w, \widehat{w} \geq 0$, the denominator is at least λ, and

$$|\widehat{s}_\lambda - s_\lambda| \leq \frac{|\delta_\ell| + r|\delta_w|}{\lambda} + \frac{|m_r| b |\delta_w|}{\lambda^2}.$$

Its derivatives satisfy

$$\frac{\partial s_\lambda}{\partial w} = -\frac{r\lambda + b\ell}{(\lambda + bw)^2}, \qquad \frac{\partial s_\lambda}{\partial \ell} = \frac{1}{\lambda + bw} \leq \frac{1}{\lambda}.$$

Thus positive row utility removes the small-denominator singularity for fixed bounded moments. It does not remove fitting error, calibration error, or distribution shift. A floor $\max(\widehat{w}, c)$ likewise bounds denominator sensitivity, while changing the unregularized population ranking.

Fixed classifiers: the unit of abstention matters. For ordinary binary precision, let $h(X) \in \{0, 1\}$ be fixed, $W = h(X)$, and $L = h(X)\mathbf{1}\{Y = 0\}$, with $T = \Pr(h(X) = 1) > 0$. An unrestricted gate observing $h(X)$ cannot exhibit strict optimal coverage reversal. Completing any feasible gate a by accepting every predicted-negative case leaves its risk and exposure unchanged and gives

$$a^+ = (1-h) + ha, \qquad c(a^+) = 1 - T + Td_W(a).$$

Whenever a positive-exposure feasible gate exists, $\sup c = 1 - T + T \sup d_W$: every row optimum is exposure-optimal, and an exposure optimum can be completed to a row optimum. This also holds with a common minimum row coverage.

A fixed *multi-label* classifier with whole-example abstention does exhibit reversal, because cases can contain different numbers of positive predictions. Take two labels and three fully observed contexts:

	K	U	V
$\Pr(X)$	$2/5$	$2/5$	$1/5$
$h(X)$	$(1,0)$	$(1,0)$	$(1,1)$
$w = \#$positive predictions	1	1	2
$\ell = \mathbb{E}[\#\text{false positives} \mid X]$	0	$3/10$	$1/2$

At K, $Y = (1,0)$; at U, $Y = (0,0)$ with probability $3/10$ and $(1,0)$ otherwise; at V, $Y = (1,0)$ or $(1,1)$ equiprobably. The gate observes all of X and independently accepts the entire prediction vector with probability $a(X)$. The risk is one minus accepted micro-precision, with $r = 1/10$ and

$c_0 = 7/20$. Every optimum accepts K; the remaining budget is $4a_U + 3a_V \leq 2$. The unique row and exposure optima are

	a	(c, d_W)	R
rows	$(1, 1/2, 0)$	$(3/5, 1/2)$	$1/10$
positive exposure	$(1, 0, 2/3)$	$(8/15, 5/9)$	$1/10$

Hence $\Delta c = 1/15$ and $\Delta d_W = 1/18$. Dividing both L, W by two gives $0 \leq L \leq W \leq 1$ without changing these results. Exact vertex enumeration and an independent LP verify both optima. The example uses whole-case acceptance; independently gating each label would define a different decision problem.

Bounded exposure limits the reversal. Suppose $0 < u \leq w(X) \leq v < \infty$. For any row and exposure optima under the same feasible set, put $x = c(a_R) - c(a_W) \geq 0$, $y = d_W(a_W) - d_W(a_R) \geq 0$, and $D = (v-u)/(v+u)$. Then

$$y \leq \frac{D - x}{1 - Dx}, \qquad 0 \leq x, y \leq D.$$

To see this, let $b = \mathbb{E}(a_R - a_W)_+$ and $a = \mathbb{E}(a_W - a_R)_+$. Then $b = a + x$, $a + b \leq 1$, and bounding the positive and negative exposure masses gives $y \leq \{(v-u)a - ux\}/\{u + (v-u)a\}$, which increases with a. Substitute $a \leq (1-x)/2$; a row floor further gives $a \leq 1 - c_0 - x$. Thus whole-example micro-precision with $1 \leq w \leq m$ has each gap at most $(m-1)/(m+1)$, equal to $1/3$ for two labels. The full supplementary proof gives the sharper floor-dependent envelope and fixed multi-label classifier constructions approaching it with unique optima; it does not claim endpoint uniqueness. Variation in w is necessary for strict reversal on this positive support, but not sufficient for a given loss law: with $\ell = 0$ throughout, accept-all optimizes both objectives for any w.

At least three contexts are needed. With unrestricted randomized acceptance under only the pooled risk cap and a common row floor, any feasible problem supported on at most two contexts, each with positive exposure, has a common optimum: fully accept the nonpositive-excess contexts, then accept the sole remaining positive-excess context as far as feasible. Strict reversal therefore requires at least three such contexts. A short analytic proof is included in `docs/two_context_no_reversal.tex`; additional constraints or restricted policy families fall outside this statement.

Additional WAPE example. The supplement provides a deterministic three-context calculation in `docs/wape_worked_example.tex`. Its row- and demand-optimal coverages are $(73/80, 11/46)$ and $(17/40, 7/23)$ at cap $1/2$; `verify_objective_geometry.py` checks the fractions and efficient segment.

A.3 Censored-tail nonidentification and bounded-tail envelope

For natural demand D, observe sales $S = \min(D, C)$ and a censoring flag, where C is an availability bound. In a mechanically censored benchmark, the retained pre-censoring target can be used for training and auditing; in natural stockouts it is generally unavailable.

Proposition 4 (Unbounded censored tails prevent a universal WAPE claim)**.** *Fix a measurable acceptance probability based on observed information and a bounded nonnegative forecast. Assume the availability bound is nonnegative and integrable and an integrable demand completion exists. Suppose the expected acceptance weight on censored cases is positive and the compatible model class places no upper bound on their demand tail. There exist observationally indistinguishable finite-mean demand completions for which accepted WAPE approaches one. Consequently, observed censored sales alone cannot uniformly establish any contract with $r < 1$ over this model class.*

This statement does not exclude identification under additional structural assumptions. With a valid known bound $D \in [C, U]$ on a censored case, convexity gives the conservative excess envelope

$$|D - f| - rD \leq \max\{|C - f| - rC,\ |U - f| - rU\}. \tag{7}$$

Inventory availability C is not itself a demand upper bound U. A maximum observed in the development sample is also insufficient to justify a population support bound.

Let $O = (X, S, C, \Delta)$ have the fixed observable law, where $C \geq 0$, $\mathbb{E}C < \infty$, and $\Delta = 1$ means $D > C$. Fix $a(X) \in [0, 1]$ and $0 \leq f(X) \leq F < \infty$, and let D_0 be an integrable compatible completion. Put $B = \{\Delta = 1\}$ and suppose $k = \mathbb{E}[a\mathbf{1}_B] > 0$. For $M > F$, replace D_0 on B by $C + M$. This preserves $S = \min(D, C)$ and Δ, and each completion remains integrable. Its accepted numerator and denominator are

$$\mathbb{E}[a|D_M - f|] = Mk + A_0, \qquad \mathbb{E}[aD_M] = Mk + B_0,$$

where the finite constants are

$$A_0 = \mathbb{E}[a\mathbf{1}_B(C - f)] + \mathbb{E}[a\mathbf{1}_{B^c}|D_0 - f|], \qquad B_0 = \mathbb{E}[a\mathbf{1}_B C] + \mathbb{E}[a\mathbf{1}_{B^c} D_0].$$

Consequently $R_M = (Mk + A_0)/(Mk + B_0) \to 1$, violating any fixed $r < 1$ for sufficiently large M. For actual randomized acceptance, introduce $U \sim \mathrm{Uniform}(0, 1)$ independent of (O, D_M) and let $A = \mathbf{1}\{U \leq a(X)\}$. Since $\mathbb{E}[A \mid O, D_M] = a(X)$, the weighted expectations equal realized-acceptance expectations. No independence of X, C, Δ is required. The construction excludes rules with zero accepted censoring weight.

If instead a valid bound gives $D \in [C, U]$, the function $z \mapsto |z - f| - rz$ is convex. Writing z as a convex combination of the endpoints bounds its value by their maximum, proving equation (7). Positive accepted demand and the coverage requirement remain separate conditions.

Standard bounded-data calibration. For a finite family fixed independently of i.i.d. calibration clusters, known bounded cluster excess and coverage permit a simultaneous Hoeffding check (Hoeffding, 1963). The supplement's `docs/standard_calibration.tex` gives the full statement and proof, and the numerical record retains 400 synthetic trials. The unbounded, dependent retail observations do not meet those assumptions; no such guarantee is invoked here.

B DATA, FITTED OBJECTS, AND EVALUATION BOUNDARIES

B.1 PARTITION, CHRONOLOGY, AND IMPLEMENTATION

The public M5 release (University of Nicosia, 2020) supplies recorded sales, calendar, and prices. Item identifiers are sorted by SHA-256 of a fixed salt, a null byte, and the identifier. The first 1,829 items define development, the next 610 the consumed guardian, and the last 610 the external test; all stores for an item stay together. Shared calendar/store shocks remain possible. For mechanical censoring, initialize the availability state to mean sales on the first 56 days, leave those days uncensored, and then update it using preceding observed sales before applying equation (4). Warmup availability is $\max(Y, 1)$; the artificial flag means $Y > C$.

Table 3: Origin-purged development targets. Guardian readout uses the same A/B/shadow dates without recalibration.

Role	Retained days	Days
Point fit / point selection	365–1313 / 1317–1433	949 / 117
Risk fitting	1436–1553	118
Calibration A / B	1555–1673 / 1674–1793	119 / 120
Point / selective shadow	1794–1913 / 1800–1913	120 / 114

Target-date calendar and price schedules are assumed known at issue time. Beyond the settings in the original selector design in Section 4, both learners use minimum leaf size 100, four threads, no early stopping, and seeds 20260906–20260908; L2 penalties are 2 for point forecasts and 3 for risk heads. Point controls share sampled rows/features, except that the uncensored-only control removes censored rows.

The 21 risk inputs summarize fixed forecasts, observed error history, horizons, observed sales/censoring, seasonal lags, and category/store identifiers. The full list and observed-error recursion are in `docs/supervised_controls.tex`; future-observation perturbations test origin validity.

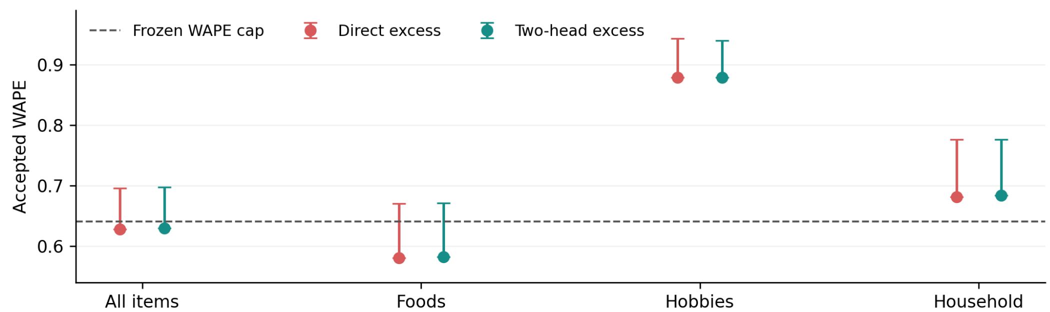

Figure 4: Previously held-out items, first prespecified seed and frozen pooled thresholds. Dots are shadow WAPE; whiskers extend to the one-sided bootstrap upper bound at nominal tail probability $0.05/48$. Hobbies and Household fail the cap even at their point estimates. All displayed adjusted upper bounds exceed the cap.

The supervised selectors retain the original $(2, 2)$ adapter, while the likelihood extension uses plain L1. The original and expanded adapter grids and secondary controls are listed in `docs/supervised_controls.tex`.

The numerical supplement retains the nine point-model controls, seed variability, and all empirical threshold frontiers. Searches use common A/B thresholds, retain ties, and reject zero demand denominators. The initial freeze uses the first coverage maximizer; the later objective audit uses the largest. Missing thresholds imply all-false masks. Development item intervals use 3,000 draws and hold fits fixed; a separate 17-origin bootstrap probes temporal dependence.

B.2 The single frozen guardian comparison

Before its one-time access, the first item-holdout comparison fixed 19 source/model objects, seven scores, two threshold modes, three seeds, the $(2, 2)$ adapter, cap 0.640129832706689, and the first-seed contrast. The 610 items comprise 283 Foods, 121 Hobbies, and 206 Household, each in ten stores. Observed histories form origin-valid features; evaluated labels span the A/B and later windows in Table 3. No learner or threshold is refitted. The original requirement demanded simultaneous risk and coverage checks across groups and periods; the comparison measures policy differences even when those operating checks fail. Machine-readable freeze and access records preserve exact timestamps, hashes, software versions, and development replay checks.

The primary uncertainty family has two pooled methods, three periods, four strata, and two endpoints: 48 bounds. Each tail receives $0.05/48$ with 10,000 item-bootstrap draws at seed 20260906. These approximate bootstrap bounds are not exact concentration guarantees. The prespecified direct-minus-two-head shadow row-coverage contrast has a separate paired 95% interval. Demand/WAPE contrasts are secondary summaries.

Table 4: Pooled endpoints from the prespecified first-seed guardian audit. Each bound uses tail probability $0.05/48$ and 10,000 item-bootstrap draws. The full 24-row, 48-endpoint record remains in the supplement; all coverage lower bounds exceed 0.35 and all WAPE upper bounds exceed 0.64013.

Method	Period	Coverage	WAPE	Cov. lower	WAPE upper
Direct	A	0.8914	0.6399	0.8745	0.7161
Direct	B	0.8794	0.6495	0.8630	0.7225
Direct	Shadow	0.8725	0.6282	0.8560	0.6959
Two-head	A	0.8895	0.6401	0.8729	0.7160
Two-head	B	0.8763	0.6504	0.8596	0.7236
Two-head	Shadow	0.8746	0.6296	0.8587	0.6971

Table 5: All 48 external secondary bounds. Each pair is row-coverage lower bound (%) / WAPE upper bound; each tail uses $0.05/48$. Every row bound passes 35%; every WAPE bound fails 0.64013. The two main directional bounds are a separate family.

Period	Stratum	$\lambda = 1$	$\lambda = 0.25$
All 23 days	All	83.90 / 0.6896	80.81 / 0.6863
	Foods	78.70 / 0.6634	76.73 / 0.6684
	Hobbies	75.58 / 0.9171	76.99 / 0.8864
	Household	91.57 / 0.7813	84.64 / 0.7618
First 11 days	All	83.10 / 0.6908	80.56 / 0.6896
	Foods	77.30 / 0.6637	76.12 / 0.6727
	Hobbies	75.72 / 0.9161	77.02 / 0.8803
	Household	90.98 / 0.7824	84.51 / 0.7642
Last 12 days	All	84.48 / 0.6907	81.00 / 0.6860
	Foods	79.78 / 0.6658	77.08 / 0.6643
	Hobbies	75.55 / 0.9201	76.84 / 0.8937
	Household	91.99 / 0.7791	84.62 / 0.7637

Across all four strata, coverage lower bounds range from 0.8040 to 0.9288 and WAPE upper bounds from 0.6704 to 0.9468. Thus every WAPE endpoint misses the cap, including Foods; this is a failure of the specified joint check, not merely the pooled rows in Table 4. A truncated derived cache was restored from unchanged models; readable arrays matched exactly, and saved masks reproduce all 504 policy/period/stratum records. Frozen results and hashes are unchanged. Subsequent objective, floor, capacity, loss, and partial-label diagnostics reuse this consumed holdout and remain exploratory.

B.3 The separately frozen external comparison

At 14:29 UTC on 6 September 2026, an initial pending plan froze direct excess versus demand-normalized error with floor 0.25. At 14:47 UTC, after inspecting the consumed first-holdout mixed-utility diagnostics, we replaced that pair with composed excess ($\lambda = 1$) versus mixed utility ($\lambda = 0.25$). No external data had been accessed under either plan. The earlier plan is preserved byte for byte; the final plan records its hash and replacement reason. The first external access was at 15:10 UTC and evaluation completed at 15:13 UTC. This is a prospectively fixed external comparison selected after first-holdout exploration, not a test independently specified before that exploration.

The final plan fixes 27 dependencies, seed 20260906, the point forecast, 220-tree heads, and development thresholds and scale. No external fit, threshold search, or scale update occurred. Exact values, timestamps, and hash-linked plan history are supplied in the reproduction package.

The 610 external items (269 Foods, 117 Hobbies, 224 Household) yield 140,300 primary rows on days 1919–1941, with every forecast origin after day 1913. Observed history through day 1913 and subsequent observations available at each origin form the features; truth is used only for evaluation. Targets 1914–1918, issued at origin 1911, enter only a prespecified 28-day descriptive sensitivity. Paired item-bootstrap draws (10,000; seed 20260906) test mixed-minus-row demand gain and row loss with one-sided 97.5% bounds. Both must exclude zero in the specified direction. Separately, two policies, three periods (1919–1941, 1919–1929, 1930–1941), four strata, and two bounds give 48 secondary endpoints, each at tail level $0.05/48$. The primary and secondary families each use nominal 0.05; they do not form one joint 95% claim. Item resampling retains stores and dates but not shared-shock uncertainty. Because the first holdout failed its operating requirements, this test separately asks whether the fixed coverage contrast transfers and whether the operating constraints hold.

The full-28-day sensitivity has row/demand coverage 86.14%/82.87% for $\lambda = 1$ and 82.51%/87.94% for $\lambda = 0.25$, with WAPE 0.6128/0.6133. It is descriptive only. The archived execution has one opening, zero fits, and zero threshold searches; saved item-level accepted counts, error sums, and demand sums reproduce the primary bounds and all 48 secondary endpoints.

Table 6: Descriptive budget accounting for the two frozen external policies, days 1919–1941. K is their intersection; U is accepted only at $\lambda = 1$, and V only at $\lambda = 0.25$. Demand shares use the total 209,893 units. Signed excess is $E_B - rM_B$; a negative value supplies slack. This analysis reuses saved predictions after the confirmation result.

Set	Rows	Demand (%)	WAPE	Mean f	Excess
Common K	109,985	79.86	0.5899	1.112	-8,421.10
Row-only U	10,557	3.32	1.0046	0.122	2,542.83
Mixed-only V	5,855	8.48	0.7794	1.703	2,480.43

Table 7: Strong fixed forecaster; three matched risk-training seeds. Values are development-shadow means $\pm$ training-seed SD (not independent-test uncertainty). Thresholds use the same two calibration blocks and row-coverage objective. Numbers in parentheses denote total boosting trees. The two-head budget-220 control uses 110 trees per head.

Score	Row coverage (%)	Demand coverage (%)	WAPE
Forecast-normalized	80.22 ± 0.27	86.58 ± 0.09	0.6216
Demand-normalized	74.17 ± 0.85	89.01 ± 0.20	0.6260
Forecast excess	81.47 ± 0.41	70.71 ± 0.59	0.6222
Two-head excess (440)	86.35 ± 0.08	81.32 ± 0.21	0.6268
Two-head excess (220)	86.61 ± 0.12	81.88 ± 0.29	0.6269
Direct excess (220)	86.30 ± 0.16	80.86 ± 0.11	0.6245

Table 8: Explicit objective-based family selection among the seven original scores, followed by threshold application to shadow periods. Three-seed means. Both objectives retain WAPE cap 0.64013 and row floor 0.35 in calibration A/B. Guardian rows are post-hoc reuse of the same consumed holdout, not new confirmation.

Cohort	Objective	Selected score	Rows (%)	Demand (%)	WAPE
Design	Row	Direct excess (220)	86.30	80.86	0.6245
Guardian	Row	Direct excess (220)	87.37	83.29	0.6280
Design	Demand	Demand-normalized	74.17	89.01	0.6260
Guardian	Demand	Demand-normalized	73.62	88.23	0.6219

Descriptive external budget accounting. We apply equation (5) retrospectively to the same saved masks (Table 6). Row-only and mixed-only sets spend 30.20% and 29.45% of common slack, yet contain 3.32% and 8.48% of demand. Both retain slack; the joint operating failure remains. This adds no confirmatory endpoint, fit, threshold, or policy; the superseded pair is not evaluated.

B.4 Retained historical controls and reproduction

The numerical supplement retains the earlier weak-forecast controls and all seed/category/period reports. The prior FreshRetailNet check used observable evaluation labels, not hidden stockout demand: shadow row coverage/WAPE 0.28163/0.30484 missed both 0.30 targets, with adjusted bounds 0.25713/0.31821. Calibration B's coverage lower bound was 0.28498. This failed check supplies no natural-stockout confirmation.

The notebook checks frozen hashes before and after replaying theory, moments, and external sufficient statistics. All verification output goes to a separate directory. Experiment scripts regenerate fits from public M5. The separate numerical assets include the derived development inputs needed for the new predictor-specific selector replay; original external replay uses retained sufficient statistics.

Table 9: Previously held-out M5 items: fixed pooled thresholds, 610 items, 6,100 series, and 695,400 shadow rows. Three-seed means; all fits and thresholds were frozen before opening. Seeds measure training variability, not three independent holdouts. Absolute-error selection accepts no rows and has undefined WAPE.

Score	Row coverage (%)	Demand coverage (%)	WAPE
Forecast-normalized	80.87	87.65	0.6219
Demand-normalized	73.62	88.23	0.6219
Forecast excess	82.74	74.59	0.6251
Two-head excess (440)	87.41	83.69	0.6297
Two-head excess (220)	87.67	84.18	0.6297
Direct excess (220)	87.37	83.29	0.6280

C Objective choice and supervised-head diagnostics

C.1 Architecture controls and first-holdout detail

Development accepted-set accounting. Direct excess gains 6.08 percentage points in mean row coverage over forecast-normalized error, but loses 5.71 points of demand coverage. At the first seed, their common accepted set covers 73.35% of rows and has WAPE 0.5958, leaving slack under the 0.64013 cap. The direct-only set adds 267,656 rows (12.84% of all rows), but only 5.69% of demand; its WAPE is 1.0026. The relative-only set contains 141,617 rows and 11.45% of demand, with WAPE 0.7915. Mean forecasts in the two differing sets are 0.039 and 1.498 units. In equation (5), the common set supplies 97,185.97 units of excess slack. The direct-only set consumes 60,141.11 (61.88%), versus 50,549.56 (52.01%) for the relative-only set. Both fit within the pooled budget despite their individual WAPE exceeding the cap. This accounts for the row gain through budget use and demand composition.

Architecture is distinct from acceptance utility. The original matched selector comparison provides an architectural control. Direct-excess, two-head, and matched-tree-budget two-head scores reach 86.30%, 86.35%, and 86.61% mean row coverage. Direct learning has slightly lower accepted WAPE (0.6245 versus 0.6268 and 0.6269); it does not establish a direct-learning advantage. The informative distinction is the coverage objective and resulting budget allocation, not whether excess is learned directly or composed from two heads. Full three-seed results are in Appendix C.

C.2 Detailed first item-holdout comparison

The freeze specifies seven scores, two threshold modes, three seeds, and a first-seed direct-versus-two-head primary contrast. The holdout uses the same A/B and later evaluation dates, without recalibration; no holdout outcome selects an original policy. Its shadow contains 695,400 rows on days 1800–1913.

The coverage trade-off persists (Table 9). Across three fixed fits, direct excess increases row coverage by 6.49 percentage points over forecast-normalized error while reducing demand coverage by 4.36 points. Two-head coverage is 87.41%, compared with 87.37% for direct learning, with WAPE 0.6297 versus 0.6280. At the prespecified first seed, direct minus two-head row coverage is -0.215 points (paired 95% item-bootstrap interval $[-0.413, -0.018]$ points); direct WAPE is lower by 0.00141. These are different trade-offs, not an equivalence result or a direct-learning advantage.

Operating constraints do not transfer jointly. First-seed direct and two-head shadow WAPE are 0.6282 and 0.6296 pooled, but approximately 0.879 for Hobbies and 0.682–0.683 for Household. In the holdout window matching calibration B, even pooled WAPE is 0.6495 and 0.6504, above the 0.64013 cap. All frozen category-specific policies abstain entirely on Hobbies because their development threshold is absent. The prespecified 48-endpoint family covers two primary methods, three periods, four strata, and two bounds, using 10,000 item-bootstrap draws. Every coverage lower bound exceeds 0.35, but every adjusted WAPE upper bound exceeds the cap; shadow pooled upper bounds are 0.6959 and 0.6971. This fails the specified joint operating check. Appendix B.2 reports all bounds and the access record.

C.3 Nested paths, family choice, and denominator regularization

For a fixed score, $A_t = \{s \le t\}$ is nested. Both $P(A_t)$ and $\mathbb{E}[Y\mathbf{1}_{A_t}]/\mathbb{E}Y$, including their empirical blockwise minima, are nondecreasing in t. WAPE need not be monotone, but the largest member of any nonempty finite feasible grid maximizes either coverage. An empty grid returns no policy.

Across 27 score/seed pairs (nine rankings, three fits), changing only the coverage objective leaves all thresholds unchanged: 24 finite and three infeasible. Family choice instead selects direct excess for rows and demand normalization for demand. The later audit uses the largest maximizer, changing 1–3 shadow rows in four configurations relative to the original first-maximizer convention; frozen results are preserved.

`CAP_REFERENCE.json` reproduces the legacy development-selection WAPE 0.7530939208313988 and retained cap, as described in Section 4. Table 10 varies the unit-sales denominator regularizer without guardian-based selection. Demand normalization wins the family comparison at floor 0.25; the smaller-floor reversals limit any claim about estimated e/μ.

Table 10: First-seed denominator-floor sensitivity on consumed guardian shadow. F/D denote forecast/demand normalization. Thresholds use development A/B; no floor is selected on guardian. Full development metrics remain in the numerical supplement.

Floor	F demand (%)	D demand (%)	F WAPE	D WAPE
0	87.77	87.53	0.6195	0.6186
0.01	87.75	87.53	0.6193	0.6186
0.05	87.97	87.53	0.6201	0.6186
0.1	87.98	87.58	0.6206	0.6188
0.25	87.73	88.10	0.6219	0.6211
0.5	85.53	89.11	0.6249	0.6257
1	74.47	86.32	0.6286	0.6297

C.4 Mixed utility and the zero-floor comparison

For a nonbinding row floor, $w_\lambda = \lambda + (1-\lambda)\mu/T$ defines a probability measure since $\mathbb{E}w_\lambda = 1$. At fixed $U_\lambda = \mathbb{E}[aw_\lambda]$, the exchange proof orders m_r/w_λ and minimizes excess. Largest feasible weighted prefixes maximize that utility; boundary randomization handles ties. At $\lambda = 0$, reject $\mu = 0, e > 0$ without losing utility; zero-error/zero-demand contexts are neutral. With a binding row floor, a finite-policy LP has Lagrangian coefficient $w_\lambda + \zeta - \tau m_r$, $\zeta \ge 0$. If $\tau > 0$, its score is $m_r/(w_\lambda + \zeta)$, with $\zeta(c - c_0) = 0$. Thus the simple score alone need not solve a binding-floor problem.

The exploratory grid uses first-seed 220-tree heads, zero-clipping, and development A/B mean demand $\hat{T}$. A/B outcomes alone select thresholds. All five guardian policies are reported; each fails period-B pooled WAPE. At $\lambda = 0$, positive-error/zero-demand rows rank last and $0/0$ is zero. The mixed $\lambda = 0.25$ point exceeds $\lambda = 0$ in both coverages; imperfect fitted heads permit this empirical dominance. Without a positive floor, forecast-normalized guardian row/demand coverage is 44.62%/87.77% (WAPE 0.6195), versus demand-normalized 62.74%/87.53% (0.6186). The latter masks equal the $\lambda = 0$ endpoint, as required by their affine score relationship.

Table 11: Fixed mixed-utility grid on consumed guardian predictions (one predeclared fit). All choices are made on development A/B; no guardian-best mixture is selected.

λ	Shadow rows (%)	Shadow demand (%)	Shadow WAPE	B WAPE
0	62.74	87.53	0.6186	0.6428
0.25	84.31	88.67	0.6272	0.6467
0.5	86.42	87.87	0.6287	0.6473
0.75	87.28	86.24	0.6292	0.6488
1	87.46	83.78	0.6296	0.6504

Table 12: Development mixed-utility path before item transfer. One common A/B-selected threshold per λ; first-seed heads. Coverages are percentages. All A/B pooled point constraints pass; these reused calibration outcomes give no independent risk guarantee.

	Calibration A			Calibration B			Later evaluation		
λ	Rows	Demand	WAPE	Rows	Demand	WAPE	Rows	Demand	WAPE
0	71.68	89.85	0.6401	67.45	88.45	0.6381	64.22	88.43	0.6231
0.25	85.25	88.67	0.6396	84.06	87.97	0.6401	83.77	88.54	0.6278
0.5	86.71	86.91	0.6388	85.69	86.57	0.6401	85.63	87.02	0.6278
0.75	87.54	84.68	0.6387	86.46	84.45	0.6401	86.36	84.68	0.6277
1	87.83	81.56	0.6390	86.63	81.46	0.6401	86.42	81.56	0.6275

C.5 Error-head bottleneck and finite-capacity effects

Table 13: Hobbies outcome substitutions, first seed. Each path optimizes its own later-window threshold, using realized labels in the excess score. Coverages are percentages; dashes mean no feasible prefix. These are samplewise diagnostics, not deployable conditional-mean oracles.

	Development			Item holdout		
Replaced head	Rows	Demand	WAPE	Rows	Demand	WAPE
Neither	0.00	0.00	–	0.00	0.00	–
Error only	76.78	23.90	0.6401	76.38	21.28	0.6401
Demand only	0.00	0.00	–	0.00	0.00	–
Both	89.15	60.94	0.6401	86.97	54.04	0.6401

The outcome-substitution diagnostic (Table 13) localizes a samplewise error-information gap. Positive-error/zero-demand rows rank last; zero-error/zero-demand rows cost no excess. Realized-label access includes irreducible noise and is not a conditional-learning guarantee.

Three 220-tree error-head interventions hold the point forecast and demand head fixed. Two Hobbies specialists share 107,384 rows from the original sample; their 21-versus-29-feature contrast holds the training population fixed. A new shared 29-feature head uses exactly the original 600,000 rows, resolving sample-size confounding for feature addition relative to the shared 21-feature head. Added features are positive-day count, positive-size mean/CV^2, and recency over 56/112 days ending at the origin. Protocols precede fitting; shared sample indices, the Hobbies subset, and original forecasts reproduce bit for bit. Future perturbations leave features unchanged. The shared 29-feature head improves both reported MSEs but yields no feasible A/B threshold or later-block prefix (Table 14). Training still uses pre-censoring error labels; no new head is evaluated on either holdout. No matched Hobbies point-forecast intervention is available, so attribution to the point model remains unresolved. Capacity controls show that enlarging only the error head lowers held-out row coverage by 0.45–0.63 points, whereas enlarging only demand raises it by 0.20–0.38 points and raises WAPE. Both heads together increase excess MSE from 1.0630 to 1.0868. The numerical supplement retains all 288 error/demand/cross-moment decompositions.

C.6 Demand-head loss controls

Squared, Poisson, and Tweedie losses target the conditional mean under their respective moment assumptions; the supplement gives the derivatives. Six matched 220-tree Poisson/Tweedie ($p = 1.5$) demand heads improve held-out demand coverage, but period-B WAPE remains 0.6470/0.6467. Full seed results are archived.

C.7 Group-budget control in a fixed learned-score class

For each category, predicted error and demand from the first-seed 220-tree heads define an 8×8 quantile grid using development A/B scores only. Bin probabilities $z_j \in [0, 1]$ are shared across both blocks. The LP maximizes minimum block row coverage, enforcing $\sum_j z_j(E_{gjb} - rM_{gjb}) \leq 0$ and $\sum_j z_j N_{gjb} \geq 0.35 N_{gb}$ separately for each group g and block b. No category borrows another's

Table 14: Development Hobbies interventions, first seed. The two shared heads use the same 600,000 rows; the two specialists use the same 107,384 Hobbies rows. All use 220 trees, the same point forecast and demand head. MSE uses the later development block. No common A/B threshold, or outcome-informed later-block prefix, meets the cap and 35% row floor. Empty-policy WAPE is undefined.

Error head	Train rows	MSE	Positive MSE	Rows (%)	WAPE	Prefix (%)
Shared, 21 features	600,000	2.3554	7.8403	0.00	–	0.00
Shared, 29 features	600,000	2.3370	7.7650	0.00	–	0.00
Hobbies, 21 features	107,384	2.4232	7.9048	0.00	–	0.00
Hobbies, 29 features	107,384	2.4399	7.9392	0.00	–	0.00

Table 15: First-seed NB grid on development data. A/B/shadow pairs are implied / realized WAPE; rows are shadow coverage. The star selects observed validation NLL among $\{1, 1.5, 2, 3, 5\}$; 0.5/10 are descriptive anchors. Intervals are 10^{-3} NLL differences from shape 2, with four-comparison Bonferroni adjustment using 10,000 paired item-bootstrap draws. They reuse the selection data and condition on fitted models. Dashes in risk columns mean no admissible threshold.

κ	Valid. NLL	ΔNLL interval	Calibration A	Calibration B	Shadow	Rows (%)
0.5	0.7541	–	– / –	– / –	– / –	0.00
1	0.7361	[2.934, 6.334]	– / –	– / –	– / –	0.00
1.5	0.7324	[0.220, 1.634]	– / –	– / –	– / –	0.00
2 *	0.7314	reference	0.6401 / 0.5597	0.6376 / 0.5578	0.6365 / 0.5484	48.74
3	0.7322	[-0.008, 1.592]	0.6401 / 0.6421	0.6368 / 0.6412	0.6355 / 0.6311	79.48
5	0.7347	[1.606, 5.031]	0.6401 / 0.6861	0.6361 / 0.6856	0.6334 / 0.6748	99.41
10	0.7392	–	0.6076 / 0.6902	0.6033 / 0.6888	0.5997 / 0.6790	100.00

budget. Calibration outcomes determine the probabilities; bins, coefficients, and probabilities are fixed before application to the consumed guardian. These are exploratory point constraints, not independent calibration bounds.

All 28 occupied Hobbies bins in A and 29 in B have positive excess. Their minimum WAPE is 0.7289866 and 0.7503685, respectively, so every nonempty fractional mixture has WAPE above the cap in either block. At the row floor, a dual certificate bounds the minimum worst-block mean excess below by 0.0142589987 units per original row. The supplement supplies coefficients and a verifier checking residual-corrected dual bounds. This infeasibility concerns the specified grid, not arbitrary features or acceptance rules.

Foods and Household attain minimum calibration row coverage 94.60% and 76.76%. Their transferred guardian B WAPE is 0.6392 and 0.6459; shadow WAPE is 0.6111 and 0.6455. Thus the group constraints expose Hobbies failure and retain Household transfer failure. Fractional metrics are ratios of expected accepted sums; a realized randomized deployment can fluctuate around them.

D Observed-sales and capacity likelihood controls

D.1 Observation model and available information

This development-only experiment keeps the plain observed-sales L1 forecast fixed; forecast choice and the cap inherit outcome-informed development history. The strict benchmark flag $\Delta = \mathbf{1}\{Y > C\}$ differs from the observable capacity hit $H = \mathbf{1}\{S \geq C\}$, which reveals only $Y \geq C$. The capacity branch uses H in both responses and historical features. Each fit samples 600,000 risk rows from days 1436–1532, validates on 1534–1553, and uses 34 origin-valid inputs, 220 trees, 31 leaves, and the common training settings. Three seeds are retained; shape selection uses observed-data validation likelihood.

Table 16: Sales-and-capacity NB $\kappa = 2$ at its observed-likelihood-selected, model-implied threshold. Bias is $(\widehat{E}/E - 1)$ or $(\widehat{M}/M - 1)$ on the same accepted rows. Each entry averages three fit-specific ratios. Positive error-mass bias exceeds positive demand-mass bias, making the implied ratio conservative in aggregate.

Period	Rows (%)	Implied WAPE	Realized WAPE	Error bias (%)	Demand bias (%)
A	49.96	0.6401	0.5568	+18.82	+3.35
B	49.44	0.6376	0.5562	+21.65	+6.12
Shadow	48.17	0.6365	0.5461	+20.59	+3.46

Let $p_\theta(y \mid X)$ be Poisson with mean μ_θ, or NB with mean μ_θ and fixed shape κ (variance $\mu_\theta + \mu_\theta^2/\kappa$). The two negative log likelihoods are

$$\ell_\Delta = -(1-\Delta)\log p_\theta(S \mid X) - \Delta \log \Pr_\theta(Y > C \mid X),$$
$$\ell_H = -(1-H)\log p_\theta(S \mid X) - H \log \Pr_\theta(Y \geq C \mid X).$$

For integer counts, the latter survival is evaluated at $C-1$. These likelihoods require an appropriate conditional count family and ignorable censoring given modeled information. Dependence of capacity on past sales does not automatically justify ignorability for a finite feature vector. The method therefore models the hidden tail; it does not nonparametrically identify it.

D.2 Implied moments, thresholding, and results

For $k = \lfloor f \rfloor$ and nonnegative f, the distribution implies

$$\hat{e} = \mathbb{E}_\theta|Y - f| = \hat{\mu} - f + 2\{f F_\theta(k) - \mathbb{E}_\theta[Y \mathbf{1}\{Y \leq k\}]\}.$$

The truncated moment is $\hat{\mu} F_{\mathrm{Pois}(\hat{\mu})}(k-1)$ for Poisson and $\hat{\mu} F_{\mathrm{NB}(\kappa+1,p)}(k-1)$ for NB, with $p = \kappa/(\kappa + \hat{\mu})$. Direct summation and finite differences check moments and likelihood derivatives, including extreme tails.

A/B thresholds maximize common row coverage under $\sum a\hat{e}/\sum a\hat{\mu} \leq r$ and the 0.35 floor. Both excess and demand-normalized rankings are archived. Implied and realized risk differ through

$$R(a) - r = \frac{\mathbb{E}[a(\hat{e} - r\hat{\mu})] + \mathbb{E}[a(e - \hat{e})] - r\mathbb{E}[a(\mu - \hat{\mu})]}{\mathbb{E}[a\mu]}.$$

Hence passing the implied constraint is not an observed-risk certificate. Table 16 reports both numerator/denominator pairs.

Across three fits per observation branch, validation selects $\kappa = 2$; Hobbies WAPE remains 0.8018. All 33 retained fits and seven derivative, moment, origin, and equality tests are archived.

D.3 Fine dispersion grid and identical-forecast comparison

All fine-grid fits share the sample, features, and forecast; protocols and hashes precede fitting. The count-matched supervised head uses the same inputs and a realized-excess target. Outcome-independent boundary draws agree with fractional-tie expectations (264 records and 21 count matches verified). Uniform fractional abstention at 48.17% instead has equal demand coverage and WAPE 0.6790.

Validation likelihood uncertainty. All seven observed NLL means reproduce exactly. The 10,000 paired-bootstrap draws resample 1,829 validation items (365,800 rows), retaining each item's stores and dates. The four adjusted comparisons use two-sided tails $0.05/(2 \times 4)$; pointwise 95% intervals are all positive. These approximate intervals reuse the data that selected shape 2 and condition on fitted models; they neither confirm the selected winner independently nor identify censored tails. Shared store/calendar shocks remain unaccounted for. Saved item sums reproduce all intervals without fitting.

D.4 Observed-data diagnostics and their target population

Theoretical selection uses conditional loss and exposure; naturally hidden demand does not supply either without assumptions. Our observed-sales-and-capacity negative-binomial experiment illustrates the issue. Among adjusted paired item-bootstrap comparisons, shape $\kappa = 3$ is the only candidate whose validation likelihood difference from selected $\kappa = 2$ includes zero, yet it breaks both realized calibration caps that $\kappa = 2$ passes. The selected model overestimates aggregate error mass by 20.59%; its pooled point pass does not validate conditional moments. Appendix D provides the likelihood, dispersion grid, and count-matched supervised control.

Further forecasting experiments on naturally stockout-annotated FreshRetailNet test observable sales outcomes, not hidden demand. A forward-time residual learner reduces WAPE on the 8,020 non-stockout test rows from 0.32928 to 0.29201, and on all 14,000 recorded-sales rows from 0.36162 to 0.33401. These later comparisons reuse the previously opened seven-day cohort; most improvement remains without the risk feature (non-stockout WAPE 0.29289). They demonstrate useful forecast adaptation within that evaluated population, but cannot certify a demand-coverage contract on unobserved stockout demand. Appendix E records the controls and uncertainty. Thus accuracy improvement, acceptance utility, and identification are three distinct requirements of a forecasting engine.

E Accuracy improvement with newly fitted acceptance policies

E.1 A distinct retrospective protocol

The additional experiment uses only the 1,829 development items and origin-valid observations; neither historical holdout is accessed. All dates have been consumed in prior research. A protocol records the predictor identities, chronological blocks, head specification, utility grid, cap, floor, and bootstrap before the additional head fits. This order makes the computation auditable; it does not erase previous knowledge of evaluation performance.

The group-scaled predictor is the first hierarchy-feature XGBoost L1 fit from the forecasting experiments, trained on observed sales with 33 origin-valid features and seven cross-store item summaries. A separately learned observable capacity-hit probability q is available. The residual predictor is

$$f_{\mathrm{res}}(X) = \max\{f_0(X) + g(X, f_0(X), q(X)), 0\},$$

where g is L1 gradient boosting trained on retained-target residuals. Days 1314–1373 fit candidate corrections and 1374–1433 select depth in $\{2, 4\}$ and tree count in $\{50, 150, 300\}$; parent scaling is a fallback. Learning rate is 0.03 and minimum leaf sample size is 50. The selected depth-4, 300-tree correction is refitted on their union (up to 400,000 sampled rows). No accuracy parameter is retuned during the new selector comparison. The correction is conventional residual learning, not a proposed architecture.

Chronos-2 uses the archived univariate, nonnegative median forecasts, context up to 2,048, horizon seven, and no cross-series learning. Weight revision, weight hashes, and prediction receipts accompany the data. Raw backbone outputs are used; the older risk-bin recalibration, which was fitted on an overlapping block, is deliberately not used in this comparison. Foundation pretraining and task-specific supervised forecasting have different information histories; the within-predictor utility contrasts are the target.

The new demand head sees 40 history/hierarchy features and q. Each predictor-specific error head also sees that predictor's forecast. Heads use squared loss, 220 trees, 31 leaves, learning rate 0.05, minimum leaf size 50, deterministic four-thread fitting, and seed 20260908. The same 600,000 sampled training rows supply Y to the demand head and $|Y - f|$ to each error head. Both outputs are clipped to nonnegative values. Thus the acceptance experiment intentionally uses benchmark truth labels; it is not an observed-sales-only solution for naturally hidden demand.

The earliest calibration origin is 1610, after all head-training targets through 1610 have been observed. The two blocks contain 35 and 28 target days; evaluation contains 114 target days with origins at least 1799. The training sample's mean demand estimates T. For each $\lambda \in \{1, 0.25, 0\}$, all distinct finite score thresholds are searched, whole ties are retained, and the largest threshold meeting pooled WAPE and row-floor constraints in both blocks is fixed. No test outcome selects a threshold or utility weight. An infeasible search would return full abstention; all nine searches are feasible. The cap uses the displayed reference 0.85×0.7530939208, differing from the historical full-precision cap by less than 3×10^{-11}.

Table 17: New within-predictor contrasts versus $\lambda = 1$, in percentage points. Marginal 95% paired item-bootstrap intervals condition on the fits and research history; 4,000 draws, 1,829 items. The pure-demand endpoint $\lambda = 0$ is included without a denominator floor.

Predictor	λ	Δ rows	Interval	Δ demand	Interval
Group-scaled L1	0.25	-5.77	$[-6.45, -5.05]$	+5.47	$[4.34, 6.64]$
Group-scaled L1	0	-24.07	$[-25.50, -22.62]$	+6.70	$[4.92, 8.53]$
Residual L1	0.25	-3.02	$[-3.57, -2.43]$	+5.87	$[4.84, 6.88]$
Residual L1	0	-16.87	$[-18.07, -15.61]$	+7.36	$[5.78, 8.94]$
Chronos-2 (raw)	0.25	-5.35	$[-5.96, -4.69]$	+6.39	$[5.45, 7.31]$
Chronos-2 (raw)	0	-22.95	$[-24.38, -21.50]$	+8.18	$[6.74, 9.60]$

Table 18: Calibration pooled and later category WAPE for the new predictors. The cap is 0.64013. All nine policies meet both calibration point constraints; Hobbies and Household miss the later cap.

Predictor	λ	Cal. A/B	Foods	Hobbies	Household
Group-scaled L1	1	0.6401/0.6259	0.5902	0.8468	0.6839
Group-scaled L1	0.25	0.6401/0.6228	0.5903	0.8321	0.6753
Group-scaled L1	0	0.6401/0.6215	0.5966	0.8135	0.6612
Residual L1	1	0.6401/0.6252	0.5901	0.8518	0.6855
Residual L1	0.25	0.6401/0.6232	0.5894	0.8366	0.6785
Residual L1	0	0.6401/0.6233	0.5940	0.8295	0.6699
Chronos-2 (raw)	1	0.6401/0.6301	0.5909	0.8520	0.6870
Chronos-2 (raw)	0.25	0.6401/0.6259	0.5898	0.8354	0.6744
Chronos-2 (raw)	0	0.6401/0.6251	0.5967	0.8164	0.6603

Table 19: Budget decomposition for $\lambda = 1$ versus $\lambda = 0.25$ in the new retrospective comparison. Demand share uses the whole evaluation denominator; excess is absolute-error mass minus r times demand mass. Negative common excess supplies slack.

Predictor	Set	Rows	Demand (%)	WAPE	Mean f	Excess
Group-scaled L1	Common	1,594,575	77.24	0.6169	1.018	-52,237.9
Group-scaled L1	Row only	196,485	3.35	0.9966	0.076	34,812.8
Group-scaled L1	Mixed only	76,262	8.82	0.7481	1.988	27,784.7
Residual L1	Common	1,730,317	81.30	0.6218	0.977	-43,525.8
Residual L1	Row only	135,756	2.55	0.9904	0.107	26,046.8
Residual L1	Mixed only	72,762	8.42	0.7217	2.208	20,039.0
Chronos-2 (raw)	Common	1,493,529	72.28	0.6196	0.955	-43,234.8
Chronos-2 (raw)	Row only	190,032	3.14	1.0049	0.070	33,442.0
Chronos-2 (raw)	Mixed only	78,383	9.53	0.7092	2.115	19,221.3

The paired bootstrap resamples 1,829 item clusters, retaining all stores and dates, with 4,000 draws at seed 20260908. These are marginal descriptive percentile intervals conditional on the predictors, fitted heads, chosen policies, and adaptive research history. They do not include model-fitting uncertainty or shared calendar/store shocks. Item sufficient statistics reproduce all intervals; saved forecast and head arrays reproduce masks and budget sums.

E.2 What the additional forecasting controls establish

The preceding forecasting study includes fixed risk-bin recalibration, forecast-value bins, top-risk-only uplift, minimum-support fallback, separately tuned residual learning with and without q, and frozen-backbone recalibration. Its role here is to supply stronger predictors and explicit limits on interpretation, not to claim a universally superior risk coordinate.

Table 20: Retained forecasting controls. All later learned-model comparisons are retrospective. FreshRetailNet non-stockout rows condition on realized availability; all-row metrics target recorded sales, not latent demand. M5 is the first-seed later development block.

Forecast	M5	FRN non-stockout	FRN all sales
Group scale	0.67139	0.32928	0.36162
Forecast-value bins	0.67045	0.32753	0.36154
Risk bins	0.66851	0.32741	0.36033
Top-risk-only uplift	0.66843	0.32504	0.35938
Residual L1 with q	0.65956	0.29201	0.33401
Residual L1 without q	0.66532	0.29289	0.33463

On M5, bin-based recalibration adds only a modest gain over the parent and does not beat the residual learner. On FreshRetailNet, a 2,000-series cohort was fixed from training data before the original one-time seven-day evaluation. Its original risk-bin versus parent comparison has uncertainty sensitive to clustering. Later forecast-bin, uplift, and learned controls reuse this cohort; they are not additional prospective tests. The original non-stockout population contains 8,020 rows, versus 14,000 recorded-sales rows overall. Training/selection for the residual learner uses the chronological tail of the training partition (earlier two blocks for fitting; later three for selection), then refits on all five. The same six-setting grid selects depth 4 and 300 trees. The base, risk head, and categorical encoder remain fixed from earlier training.

FreshRetailNet risk-bin minus residual WAPE is 0.03541 on non-stockout rows. Paired 95% intervals are [0.02813,0.04273] by series, [0.02824,0.04368] by store, [0.01535,0.05326] by product, and [0.02135,0.04983] by date; only seven dates are available. The no-q follow-up is adaptive and separately tuned. Its WAPE difference from the full residual learner is 0.00088, with product-cluster interval [-0.00016,0.00197]. Most forecasting improvement therefore cannot be attributed to the risk feature. The corrected predictor's RMSE decreases from 0.74595 to 0.62157 and signed bias from -14.04% to -1.92% on that population. It still worsens 777 of 2,000 series. None of these observations identifies demand on stockout rows or measures inventory cost.

Frozen Chronos recalibration also improves within-backbone WAPE, but its calibration period differs from that of the main tabular predictor and its bin-control advantage changes under asymmetric pinball losses. These historical comparisons are preserved in the forecasting evidence archive. They are not pooled with the original external test or counted as independent natural-demand confirmation. Numerical dependency repairs and column-label audits are recorded there; the selected forecasts used in the present experiment reproduce the corrected reference values exactly.

F Additional cap and classification audits

F.1 Why full-coverage failure is not sufficient

If full-coverage risk is at most r, accept-all is feasible and maximizes both coverages. The converse fails even with positive exposure: take two equally probable contexts with $w = 1$, $\ell = (0, 1)$, and $r = 1/4$. Full-coverage risk is $1/2 > r$, but feasibility is $3a_2 \le a_1$. The common unique optimum is $(1, 1/3)$, with both coverages $2/3$ and risk $1/4$. Any row floor at most $2/3$ leaves this example feasible. Unlike $\ell \equiv 0$, this counterexample satisfies the premise of full-coverage cap failure.

F.2 Development cap/floor sweep and conditional-error baseline

We reuse the three predictors and their frozen demand and squared-loss error heads from Section 5.6, without fitting or accessing either holdout. The declared grid is $r \in \{.60, .62, .64012983268, .66, .68, .70\}$ and $c_0 \in \{0, .35, .60, .80\}$. For each cap we compare conditional error $\hat{e}$, row excess, mixed utility ($\lambda = .25$), and exposure ratio ($\lambda = 0$). The conditional-error comparator instantiates the regression-based uncertainty construction of Franc et al. (2023); its threshold is recalibrated to our ratio-risk constraint. This is neither a reproduction of their SELE training objective nor a claim to their original risk guarantees.

Table 21: Development sensitivity at row floor .35. Entries are mixed-minus-row case/exposure coverage differences in percentage points. Error-only reports whether the conditional-error family is feasible on calibration A/B.

Predictor	Cap	Δc	Δd	Error-only feasible
Base	0.600	-8.64	+9.22	no
Base	0.620	-7.55	+7.30	no
Base	0.640	-5.77	+5.47	no
Base	0.660	-4.85	+3.01	no
Base	0.680	-0.73	+0.19	yes
Base	0.700	+0.00	+0.05	yes
Improved	0.600	-6.46	+10.27	no
Improved	0.620	-4.68	+8.31	no
Improved	0.640	-3.02	+5.87	no
Improved	0.660	-1.69	+2.01	no
Improved	0.680	+0.00	+0.08	yes
Improved	0.700	+0.00	+0.09	yes
Chronos-2	0.600	-8.25	+9.76	no
Chronos-2	0.620	-6.49	+7.89	no
Chronos-2	0.640	-5.35	+6.39	no
Chronos-2	0.660	-5.10	+5.10	no
Chronos-2	0.680	-3.80	+1.35	no
Chronos-2	0.700	+0.00	+0.05	yes

For every ranking, we search its distinct finite calibration scores, accepting whole ties, and choose the largest threshold satisfying both chronological calibration blocks. The same rule is applied to all methods. An infeasible family accepts nothing, with undefined risk rather than risk zero. The saved grid contains 288 configurations. At the original cap and floor, all nine previously reported objective-specific thresholds and coverage/risk triples reproduce to 10^{-12}.

Table 21 shows the mixed-minus-row contrast. For the base and improved predictors it persists through .66, and for Chronos through .68. At loose caps both coverages approach one. Tiny remaining differences reflect transfer of finite observed-score thresholds, rather than an explicit accept-all candidate; they are not population reversals. Raising the floor never changes a surviving threshold: 31 grid records become infeasible relative to floor zero. This is the nested-family prediction, and clarifies that a row floor filters feasibility rather than providing an alternative optimum within a fixed ranking. All results remain retrospective and conditional on these fits.

F.3 YEAST: A PRESPECIFIED REAL-DATA NEGATIVE CHECK

We use the official Mulan Yeast train/test files (1,500/917 examples, 103 features, 14 labels; `https://github.com/tsoumakas/mulan/tree/master/data/multi-label/yeast`). Before evaluating test labels, we record the protocol, then fit independent standardized logistic classifiers on 1,125 training examples and calibrate on the remaining 375 (seed 20260909). Each classifier has $C = 1$ and a fixed .5 probability threshold. Whole-example acceptance retains or removes every predicted label together. The observed weight is the number of predicted positives, and loss is the number of false positives; thus risk is one minus micro-precision, not false negatives or true-positive coverage. Estimated loss sums $1 - \hat{p}_j$ over predicted-positive labels. Zero-weight examples also have zero loss and are accepted freely.

The primary cap is .85 times calibration full-coverage risk; .75, .95, and 1.05 are secondary declared factors. The row floor is .35. Conditional error, row excess, and exposure-ratio thresholds use the same largest-feasible whole-tie rule. Model files and thresholds are frozen before the single test-label read. The frozen classifier's test exposure ranges from zero to seven (nine zero-weight examples), and full risk is .33956.

At the primary cap .27559, both objective-specific policies accept 574/917 examples. Exposure coverage changes from 62.850% to 62.941%: $+0.090$ points with paired example-bootstrap 95% interval $[-0.181, 0.452]$. The case difference is zero, with interval $[-0.327, 0.327]$ points. Both test point risks exceed the cap. This does not establish strict objective reversal on Yeast, despite

Table 22: All prespecified Yeast test results. Cap factors multiply the calibration full risk. Coverage is in percent; risk is false positives per predicted positive.

Factor	Ranking	Cases	Exposure	Risk
0.75	Conditional error	42.86	34.05	0.2354
0.75	Row excess	37.95	36.49	0.2263
0.75	Exposure ratio	38.60	36.46	0.2223
0.85	Conditional error	67.50	59.20	0.2840
0.85	Row excess	62.60	62.85	0.2910
0.85	Exposure ratio	62.60	62.94	0.2896
0.95	Conditional error	92.15	88.76	0.3269
0.95	Row excess	86.48	86.11	0.3202
0.95	Exposure ratio	83.97	85.36	0.3187
1.05	Conditional error	100.00	100.00	0.3396
1.05	Row excess	99.67	99.61	0.3382
1.05	Exposure ratio	99.89	99.97	0.3394

varying exposure. It is consistent with the distinction between an existence theorem and outcomes of estimated scores. We retain the negative result rather than replacing the primary cap after seeing test outcomes. Table 22 reports every declared policy; secondary contrasts and all masks are in the supplement. The 2,000-resample intervals condition on the fit and are marginal, not simultaneous over the cap grid.

G Non-retail checks and additional diagnostics

G.1 Design fixed before the new evaluations

After the Yeast result, we specify two additional datasets, a single seed (20260910), models, all selectors, primary reporting cap factor .85, and secondary factors $\{.75, .95, 1.05\}$. Each reporting cap multiplies full calibration risk. Thresholds must satisfy .95 times that cap on calibration, supplying a fixed margin rather than a guarantee, and row coverage at least .35. Methods rank by predicted error, signed row excess, mixed utility ($\lambda = .25$), exposure ratio, or descending predicted weight. Every distinct calibration score is considered with whole ties; accept-all is also included when calibration permits it. Infeasible methods return an empty set and undefined risk. Results at loose reporting caps may still reject cases because calibration uses a margin.

UCI Bike Sharing (`https://archive.ics.uci.edu/dataset/275/bike+sharing+dataset`) supplies recorded hourly rentals, not latent unconstrained demand. We randomize days into 40/20/20/20 percent predictor-fit, error-head-fit, calibration, and test groups. Calendar and contemporaneous weather variables are features; counts and their registered/casual components are excluded. Fixed LightGBM models use 200 trees, 31 leaves, learning rate .05, and minimum leaf sample count 30. The point predictor uses median loss; a Poisson head estimates mean exposure. Squared-loss regression estimates absolute errors on a disjoint head-fit partition. No censoring is introduced. The 3,477-hour test has 147 days. This split tests conditional prediction on new day groups, not chronological extrapolation with forecast weather.

Delicious uses the LIBSVM redistribution of Mulan (`https://www.csie.ntu.edu.tw/~cjlin/libsvmtools/datasets/multilabel.html`), with 500 supplied features and 983 labels. A seeded 60/20/20 row split separates fitting, calibration, and testing. Independent logistic classifiers use $C = 1$, liblinear, and fixed probability threshold .5 (primary) or .2 (secondary). Weight is the predicted-positive count and loss is false-positive count. Test labels are parsed only after the model and all thresholds are frozen. At the primary cap, neither objective-specific family can meet the calibration constraint and row floor. This remains a negative result; empty-set intervals are not evidence of equivalence between useful policies.

For each dataset we retain all declared results and perform 4,000 paired resamples of test units: days for Bike Sharing and examples for Delicious. Marginal 95% intervals and 98.75% percentile intervals are supplied. The latter apply a Bonferroni adjustment over two coverage differences in each of the two primary dataset comparisons, conditional on fitted models. They are approximate

Table 23: Mobility primary test, cap $.13281$. Error-only is infeasible at calibration. Coverage is percent.

Ranking	Cases	Exposure	Risk
Conditional error	0.00	0.00	–
Row excess	87.89	85.54	0.12999
Mixed	86.54	89.66	0.12962
Exposure ratio	74.55	90.56	0.12826
Descending mean	37.79	73.66	0.12396

Table 24: Descending predicted demand at the original M5 cap; development evaluation only.

Predictor	Cases (%)	Demand (%)	WAPE
Base	45.24	88.24	0.62696
Improved	54.32	92.15	0.63058
Chronos-2	41.14	86.06	0.62997

bootstrap statements, not distribution-free risk guarantees. No model or cap was changed after test evaluation.

G.2 Mobility reversal and its budget

At the primary cap, demand-minus-row coverage is -13.34 case points and $+5.01$ exposure points. The adjusted intervals are $[-15.86, -10.77]$ and $[2.27, 8.16]$ points. Both test point risks are below $.13281$. The common set has 2,460 hours and supplies 5,219.29 loss units of slack. Row-only acceptance adds 596 hours and 15,531 rentals, costing 3,568.63; demand-only acceptance adds 132 hours and 49,847 rentals, costing 2,399.22. The union has risk $.13398$ and excess $+748.56$: these two observed additions cannot both fit the reporting cap. This decomposition is descriptive and was not a prespecified endpoint. In contrast, the original retail external sets have a feasible union; that experiment verifies a selected policy contrast rather than binding competition.

G.3 Descending-demand baseline on M5

For the original cap and row floor, we use exactly the same two-block threshold calibration as the earlier sweep, substituting $-\hat{\mu}$ for the score. Table 24 shows that this simple ranking covers more demand than the fitted pure-ratio rule in all three cases, but accepts fewer rows. This refutes a universal empirical-superiority interpretation of estimated ratio ranking; it does not contradict the exact-conditional-information theorem.

G.4 Rank agreement as a diagnostic, not a certificate

On positive estimated exposure, Kendall's τ between row-excess and ratio rankings is $.872$ for Yeast, $.300/.298/.347$ for the three M5 predictors, $.635$ for Bike Sharing, and $.760$ for Delicious's primary classifier. M5 uses a seeded 100,000-row sample; the smaller datasets use all test examples. The high Yeast agreement is consistent with similar acceptance sets, while lower agreement indicates scope for different allocations. Agreement alone does not determine thresholds, feasibility, exposure masses, or their generalization. In particular, Delicious is infeasible despite nonidentical rankings. These associations are descriptive, not an identified causal explanation or a necessary-and-sufficient diagnostic. Zero-exposure cases are excluded from the correlation and are not silently included in a positive-lower-bound envelope.

G.5 Original point-forecast adapter specifications

The reference forecaster fits histogram gradient boosting with Poisson loss to observed sales. Its expectation–maximization (EM) correction repeatedly imputes censored responses using clipped Poisson means conditional on exceeding capacity, then refits; a separate classifier predicts censoring probability. New LightGBM controls share origin-valid features, 1.2 million sampled rows, 300

Table 25: Declared secondary non-retail results: row and exposure rankings. A dash denotes calibration infeasibility, not zero risk. All five ranking families are in the accompanying JSON records.

Data/threshold	Factor	Row c	Row d	Row risk	Ratio c	Ratio d	Ratio risk
Bike	0.75	61.92	58.61	0.1124	57.95	78.85	0.1203
Bike	0.85	87.89	85.54	0.1300	74.55	90.56	0.1283
Bike	0.95	95.48	94.71	0.1398	91.54	97.64	0.1406
Bike	1.05	100.00	100.00	0.1515	99.63	99.97	0.1513
Delicious 0.2	0.75	0.00	0.00	–	0.00	0.00	–
Delicious 0.2	0.85	0.00	0.00	–	0.00	0.00	–
Delicious 0.2	0.95	0.00	0.00	–	0.00	0.00	–
Delicious 0.2	1.05	98.70	98.80	0.6380	97.76	98.65	0.6380
Delicious 0.5	0.75	0.00	0.00	–	0.00	0.00	–
Delicious 0.5	0.85	0.00	0.00	–	0.00	0.00	–
Delicious 0.5	0.95	0.00	0.00	–	0.00	0.00	–
Delicious 0.5	1.05	97.86	99.05	0.4242	94.19	98.48	0.4239

trees, 63 leaves, learning rate 0.05, and three seeds. Objectives are Poisson or L1 on observed sales, with an uncensored-row-only L1 ablation. For L1 forecast b, the adapter reuses the earlier EM prediction e_{EM} and predicted censor probability $\hat{p}_C$:

$$f_{\alpha,\gamma} = b + \alpha \hat{p}_C^{\gamma} \max(e_{\mathrm{EM}} - b, 0). \tag{8}$$

Only point selection chooses α, γ. Controls include global/category rescaling and the same ungated correction. The supervised selector comparisons retain the first point seed and $(\alpha, \gamma) = (2, 2)$. The sales-and-capacity extension uses the plain observed-sales L1 forecaster without this adapter.